

Experimental Quantification of the Phonon Drag Deformation Mechanism in Metals at Extreme Strain Rates

by

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Abstract

Extreme strain rate deformations, above 10^6 s^{-1} , are seen across many fields of science and engineering; from meteorite impacts and impact induced crystallographic phase changes to high-speed machining and additive manufacturing. Despite the range of applications, many common high-rate impact experiments are intrinsically limited to strain rates of only 10^4 s^{-1} before complicating the material deformation with a superimposing state of shock due to high impact pressures. However, recent advances in optically driven microballistics using laser induced projectile impact tests have provided a new quantitative look into extreme mechanics of materials, at rates above 10^6 s^{-1} and well below the onset of shock effects.

As deformation strain rates increase, additional strengthening mechanisms in metals become available, leading to a change in the underlying physics of dislocation motion and an increase in strength. This thesis first explores the mechanical properties of pure metals when deformed at extreme strain rates – both in ambient conditions and elevated temperatures. Using an array of complimentary characterization methods, two independent measurements of strength, the dynamic strength and dynamic hardness, are assessed. As the temperature is increased from ambient, the strength and hardness of pure metals both increase an appreciable amount. At these deformation rates, conventional thermal softening effects are now in competition with anti-thermal hardening that arises from ballistic transport of dislocations from phonon interactions in the crystal lattice. These effects are quantified systematically and it is shown that the anomalous thermal strengthening seen is, thermodynamically and kinetically, the expected form of plasticity under these impact conditions.

Next, the limits of where this anomalous thermal strengthening occur in metals are investigated. First, solute elements are added to pure Ni to evaluate how additional dislocation pinning mechanisms effect the strength at ambient and elevated temperatures during extreme strain rate deformations. The strengthen increase due to solute pinning of dislocations is additive to the other strengthening mechanisms, yet thermally controlled, which provides a transition from ballistic transport of dislocations to thermally activated strengthening at a critical concentration of solutes. Finally, the upper bound of temperature for dislocation phonon drag strengthening is assessed. While it was shown that pure metals increase strength with increasing temperature, this “hotter-is-stronger” trend breaks down as the temperature approaches the melting point of the metal. Using Sn, due to its low melting temperature, the breakdown from “hotter-is-stronger” to “hotter-is-softer” as the initial substrate temperature approaches the melting temperature is systematically explored.

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Table of contents

Abstract.....	2
Acknowledgement	3
List of figures.....	5
Chapter 1: High strain rate deformations.....	9
1.1 Dislocation motion and deformation mechanisms in metals.....	11
1.2 High-rate deformation mechanisms	13
1.3 Laser induced projectile impact testing.....	18
1.3.1 High temperature LIPIT	23
1.3.2 Using LIPIT as a mechanical testbed	24
1.4 Thesis Outline.....	29
Chapter 2: Pure metals increase strength with increasing temperature at extreme strain rates.....	31
2.1: Methods.....	33
2.2: Quantitative measurements of strength and hardness.....	34
2.3: Deformation Mechanisms	41
2.4: Implications and Conclusions.....	50
Chapter 3: At extreme rates pure metals thermally harden while alloys thermally soften	51
3.1: Extreme Strain Rate Testing on Ni Alloys.....	52
3.2: Deformation mechanisms including solute effects	59
3.3: Summary and Implications	69
Chapter 4: Breakdown of the “hotter-is-stronger” trend close to the melting temperature	70
4.1: Quantifying High-Rate Properties of Tin.....	71
4.2: Deformation Mechanisms Pre- and Post- Melting	77
4.3: Summary	80
Chapter 5: Conclusions and future work	82
References	87

List of figures

Figure 1: Many common impact testbeds and applications are plotted based on their characteristic size and impact velocity. Contours of constant strain rate are plotted as gray dotted lines. From Ref. ¹⁸	10
Figure 2: A high strain rate deformation mechanism map for Ti is shown, from Ref. ²⁵ . At the highest rates, above 10^4 s^{-1} , a drag-controlled plasticity region appears where the strength increases with increasing temperature.	15
Figure 3: The dislocation damping constant for Al, solid black circles, is plotted along with the theoretical prediction for the electron and phonon damping coefficients from Mason, dotted lines ²⁴ . The damping coefficient seen experimentally is a linear combination of these.	16
Figure 4: The theoretical prediction of the combination of thermal strengthening (labeled Eq. 4), drag strengthening (labeled Eq. 5), and the combination of the two (labeled Eq. 3). From Ref. ²⁰	17
Figure 5: The flow stress in Cu as a function of strain rate is shown up to rates of $\sim 10^5 \text{ s}^{-1}$. Interestingly, a transition in the deformation mechanism is predicted due to the upturn in strength starting around 10^4 s^{-1} . From Ref. ²⁶	18
Figure 6: A schematic of the standard LIPIT set up is shown. Ref. ²⁹	20
Figure 7: An exemplar image sequence of a 14- μm ceramic alumina microparticle impacting and rebounding off of a polished Cu substrate at 425 m/s and 40 m/s ⁴⁰	21
Figure 8: a) A schematic of the LIPIT with the glass launch pad assembly is shown. b) The standard launch pad design using the polymer layer is both schematically and experimentally shown. The expansion of the polymer bubble the accelerates the projectile forward is seen. c) The alternate glass sandwich launch pad design. Because the polymer layer is eliminated, there is no longer the expanding bubble or degradation of material when heated to several hundred degrees Celsius. Ref. ⁴²	23
Figure 9: The dynamic hardness of copper and iron measured using single microparticle impact testing is shown as a function of strain rate. When compared to studies conducted at other strain rates, a dramatic increase in hardness is seen. Ref. ⁴⁰	25
Figure 10: The velocity for the onset of erosion can be predicted for any particle substrate pairing from Eq. 9 ³⁶	27
Figure 11: The impact craters from high velocity impacts of steel particles on to tin substrates are shown for increasing impact velocities. At 310 m/s the first signatures of melting are seen. From Ref. ³⁰	28
Figure 12: The impact and rebound trajectories of three alumina particles impacting a copper substrate at 20 °C (blue), 100 °C (orange), and 177 °C (red). As the temperature is increased, the rebound velocity of the impacting particle increases, as shown by both the plotted rebound trajectories and gray insets of the particle 600 ns after rebound. Insets of plan and line scan views of each impact crater, measured with 3D laser scanning microscopy, show that as the temperature is increased the impact crater depth and width decrease.....	36
Figure 13: (a) The impact velocity versus coefficient of restitution, v_r/v_i , is plotted on a double logarithmic scale for a range of impact velocities. Across the span of impact velocities, the impacts at higher temperatures led to a faster rebound. The solid lines represent the scaling law for ideally plastic impacts with the fitting parameter, Y_d , increasing with temperature. Uncertainty in measured velocities	

is $\pm 2\%$, which is on the order of the size of the data markers. **(b)** Dynamic hardness measurements from each impact are plotted against deformation strain rate. The shaded regions serve as a guide for the eye to show where the majority of the hardness values fall for each temperature. As with strength, the hardness trends upward with increasing temperature, and measurement uncertainties are on the order of the size of the data markers. 39

Figure 14: **(a)** The impact velocity versus CoR is plotted on a double logarithmic scale for alumina impacts on gold substrates at 20 °C (blue) and 97 °C (red). Using scaling for ideally plastic impacts the dynamic yield strength for each impact temperature is calculated to be $Y_{d, 20^\circ\text{C}} = 297 \text{ MPa}$ and $Y_{d, 200^\circ\text{C}} = 311 \text{ MPa}$. Uncertainty in measured velocities is $\pm 2\%$, which is on the order of the size of the data markers. **(b)** The dynamic hardness of titanium is plotted for both 20 °C (blue) and 97 °C (red). For impact at similar strain rates, the tests at 97 °C show a higher hardness, and measurement uncertainties are on the order of the size of the data markers..... 40

Figure 15: **(a)** The impact velocity versus CoR is plotted on a double logarithmic scale for alumina impacts on titanium substrates at 20 °C (blue) and 200 °C (red). Y_d for each impact temperature is calculated to be $Y_{d, 20^\circ\text{C}} = 776 \text{ MPa}$ and $Y_{d, 200^\circ\text{C}} = 890 \text{ MPa}$. Uncertainty in measured velocities is $\pm 2\%$, which is on the order of the size of the data markers. **(b)** H_d of titanium is plotted for both 20 °C (blue) and 200 °C (red). For impact at similar strain rates, the tests at 200 °C show a higher hardness, and measurement uncertainties are on the order of the size of the data markers..... 41

Figure 16: The contribution of each strength term for pure copper, calculated for a range of temperatures at a fixed strain rate of 10^7 s^{-1} . The magnitude of the increase in dislocation drag strengthening is greater than the magnitude of the decrease in all other strength terms, leading to a net increase in the total strength with temperature at this extreme strain rate. The experimental data points, plotted in red, are in good agreement with the total strength from all a summation of each strength component. Error bars on the experimental data points are the standard deviation of the strength measurements..... 46

Figure 17: The blue and red dotted lines serve as a guide for the eye to connect hardness values, converted from strength values from Ref.²⁰ (empty circles), at low rates and dynamic hardness values from this work (filled circles). The apparent activation energy, Q_{app} , for plasticity (solid black line) of copper is also plotted as a function of strain rate at a constant temperature of 20 °C based on the model in the supplemental section. At rates below $\sim 10^5 \text{ s}^{-1}$, Q_{app} changes from positive, indicating thermally activated dislocation motion, to negative, one of the necessary conditions for phonon drag to be significant. 49

Figure 18: The impact velocity versus coefficient of restitution is plotted on a double logarithmic scale for **(a)** 99.999% Ni **(b)** 99.995% Ni **(c)** 99.95% Ni **(d)** 99.5% Ni and **(e)** 99% Ni at 20 °C and 155 °C. There is a clear change from thermal hardening to thermal softening when increasing the solute content in Ni with 99.95% Ni serving as the approximate transition point. Solid lines indicate the parabolic scaling of the impact and rebound behavior as proposed by Wu et al. This scaling was used to calculate the dynamic yield strength σ_y for each material at each temperature..... 54

Figure 19: Laser scanning confocal micrographs are shown for **(a,b)** 99.999% Ni, **(c,d)** 99.95% Ni, and **(e,f)** 99% Ni with inserts showing the measured crater volume and impact velocities. The crater volume increases from **(a)** to **(b)** indicating the expected thermal hardening in pure metals. The crater volume decreases from **(e)** to **(f)** indicating classic thermal softening in the lowest purity material. The crater volumes in **(c)** to **(d)** are approximately equal corresponding to the transition point from thermal

hardening to thermal softening. What is more, as the purity level decreases, the crater volume decreases, another expected form of plasticity where alloy elements increase the hardness of a metal.⁵⁷

Figure 20: The dynamic hardness for each impact experiment was calculated and plotted as a cumulative probability for each test material and temperature. As with the dynamic yield strength and measured crater volumes, the highest purity samples exhibited thermal hardening and the lowest purity samples exhibited thermal softening. Additionally, The average hardness at 20 °C is increasing with increasing solute content, the expected form of plasticity. 58

Figure 21: The experimentally assessed dynamic yield strength for each test material is plotted as a function of interstitial concentration, filled circles. As the interstitial concentration increases, for each temperature, the yield strength increases. Solid lines show the total yield strength predicted from the model, including the solute strengthening term from Eq. 27. As the interstitial concentration increases from 0.001%, the materials exhibit thermal hardening then cross over to thermal softening at an interstitial concentration of ~0.4%. This cross over from “hotter-is-stronger” to “hotter-is-softer” arises from a change in the dislocation kinetics from ballistic transport of dislocations to solute drag controlled, which is thermally activated. 64

Figure 22: The coefficient of restitution is plotted as a function of impact velocity for **(a)** 99.99% Ti and for **(b)** CP-Ti, at 20 °C and 200 °C. 99.99% Ti thermally strengthens while CP-Ti thermally softens due to the increased number of pinning points for dislocations from the higher solute content. Solid line insets over the data show the parabolic scaling of the impact from the plasticity model by Wu et al. The dynamic yield strengths are calculated for each sample. The dynamic hardness of each test material is plotted as a cumulative probability in **(c)**. As with the dynamic yield strength, the dynamic hardness increasing with increasing temperature for the 99.99% Ti samples and decreases with increasing temperature for the CP-Ti samples. Data for 99.99% Ti was taken from our previous work in Ref. ³⁹..... 67

Figure 23: The coefficient of restitution is plotted as a function of impact velocity for **(a)** 99.999% Au and for **(b)** 99.955 Au, at 20 °C and 97 °C. 99.999% Au thermally strengthens while 99.95% thermally softens. Solid line insets over the data show the parabolic scaling of the impact from the plasticity model by Wu et al. The dynamic yield strengths are calculated for each sample. The dynamic hardness of each test material is plotted as a cumulative probability in **(c)**. While the hardness data for 99.999% Au shows significant overlap, consistent with the coefficient of restitution plot, the trends still hold where the high purity sample is thermally hardening while the lower purity sample is thermally softening. A lower maximum test temperature was used because the Au samples were work hardened prior to impact experiments so recrystallization of the work hardened samples was avoided..... 68

Figure 24: **a)** The data for each impact are plotted as the impact velocity versus coefficient of restitution (v_r/v_i) for each of the four temperatures, data for 20 °C reproduced from ref ³⁰. **b)** Only the impacts that follow the power law scaling relation are shown. Here a clear thermal hardening trend emerges from the data, seen as both faster rebounds at higher temperatures and the calculated dynamic yield strength values, Y_d . **c)** Impacts that diverge from the power law scaling, at faster impact velocities, are plotted. The thermal hardening trend appears to change back to conventional thermal softening as impacts at higher temperatures have slower rebound velocities. Impact craters from steel particle impacts are shown at three temperatures tested: 50 °C (blue), 100 °C (orange), and 150 °C (red). For each temperature, the impact craters show signatures of only plasticity and plasticity plus melting. The scale bar in each image is 3 μ m. 74

Figure 25: The dynamic hardness as a function of strain rate is plotted for each temperature in **a)** and the non-melting regime and **b)** the melting regime. The shaded regions are guides for the eye and highlight the data below, **a)**, and above **b)** the onset velocity for melting. The hardness trend in **a)** shows clear signs of thermal hardening with the hardness values at hotter temperatures being higher than at cooler temperatures. Past the point of substrate melting, **b)**, the trend in hardness changes where the hardness values at cooler temperatures are higher than at hotter temperatures. **c)** The hardness data for 20 °C and 150 °C are plotted separately from the other data, cleanly highlighting the thermal hardening and thermal softening regimes. 76

Figure 26: Hardness color contours as a function of temperature and strain rate are plotted for tin, up to the melting temperature (505 K). The thick black line indicates the boundary between melting and no melting for an impact under a given initial temperature and strain rate, and is the mechanistic transition point between thermal hardening (below) and thermal softening (above). 80

Chapter 1: High strain rate deformations

Understanding material behavior under high strain rate conditions is important across many applications including solid state additive manufacturing ^{1,2}, erosion ³, micrometeorite impacts ⁴, mechanical alloying ⁵, and impact induced phase transformations ^{6,7}. Common test methods for studying dynamic properties of materials include impact nanoindentation, dynamic nanoindentation ⁸, Charpy impact testing ^{9,10}, freefall drop tests ¹¹, Taylor impact tests ¹², split-Hopkinson pressure bar tests ^{13,14}, gas gun impacts ¹⁵, plate impacts, laser induced forward transfer (LIFT) ¹⁶, and more recently laser-induced projectile impact tests (LIPIT) ¹⁷; with each of these test methods covers a range of strain rates, length-, velocity-, and time-scales. Figure 1, adapted from Ref. ¹⁸ summarizes several of the testbeds for studying the dynamic properties and deformations of materials across the characteristic size-velocity space for mechanics. Classical mechanical tests generally only probe macroscale material samples at low strain rates, however, many important aspects of high strain rate mechanics, such as microstructural effects and dislocation dynamics, involve nano to micro length scales at strain rates above 10^6 s^{-1} .

Many commonly used high-rate mechanical tests are intrinsically limited to strain rates of $\sim 10^4 \text{ s}^{-1}$. In order to generate higher strain rates, faster impact velocities are needed. However, there exists a limit where faster impact velocities produce strong shock wave effects in the material of interest. Impacts can produce pressures on the order of hundreds of GPa in one dimensional strain conditions and generate shock waves that travel in both the impactor and target material. These shock waves can convolute the material response and make it difficult to untangle shock effects from the mechanical response of a metal under high strain rate deformations. Microprojectile impact experiments, including those using LIPIT (detailed in

section 1.3), have overcome some of the challenges faced when trying to create extreme strain rate deformations that are free from strong shock effects. LIPIT offers the ability to produce deformation rates on the order of 10^6 - 10^9 s⁻¹ while keeping impact pressures to just a few GPa.

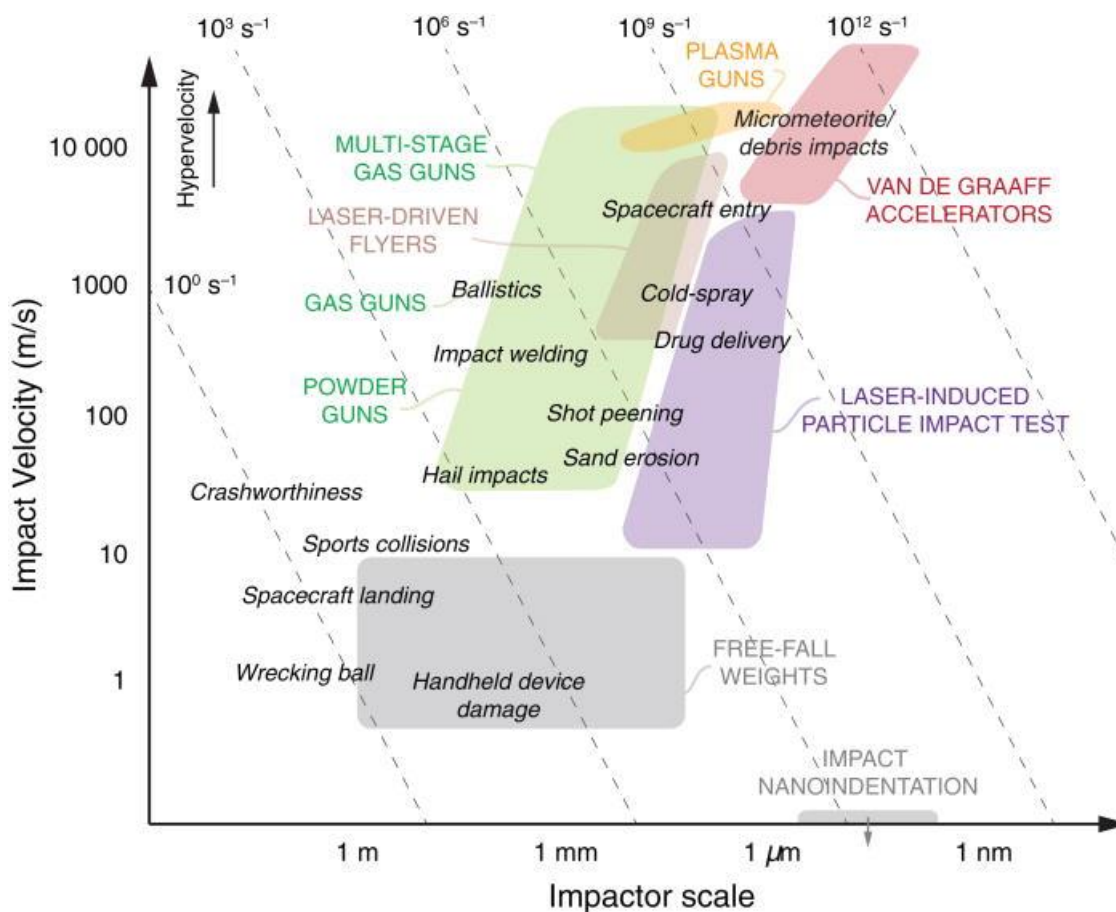


Figure 1: Many common impact testbeds and applications are plotted based on their characteristic size and impact velocity. Contours of constant strain rate are plotted as gray dotted lines. From Ref. ¹⁸

1.1 Dislocation motion and deformation mechanisms in metals

The way a metal plastically deforms is dependent on temperature, stress, strain, and/or the strain rate at which it is deformed^{19,20}. Plasticity is a kinetically driven process²¹, where atoms move past each other, dislocations glide and climb, grain boundaries slide, and twin boundaries form in metals. The rate at which these processes occur can be used to describe the dominant deformation mechanism under a given set of conditions.

One of the first attempts at constructing where the boundaries of each deformation mechanism exist, for metals, was by Ashby¹⁹, who found there to be several possible deformation mechanisms at low strain rates ($<1 \text{ s}^{-1}$): defect-less flow, dislocation glide, diffusional creep, dislocation creep, and twinning. It was noted however, that other mechanisms are possible, but unknown and underexplored due to a dearth in experimental data in some regimes.

At deformation rates up to 10^4 s^{-1} , the strength of a metal is dominated by obstacle-controlled strengthening that is, in general, thermally activated dislocation motion. The strength arises from the intrinsic lattice resistance to dislocation motion, solid solution strengthening, work hardening, and grain size strengthening. Typically, dislocations must overcome two types of barriers in order to move: short-range and long-range barriers. Short-range barriers include Peierls Nabarro barriers and solute atoms, while long-range barriers are precipitates and grain boundaries. The main distinction between these barriers is that dislocations can overcome short-range barriers by thermal fluctuations and long-range barriers cannot. Because of this, short-range barriers produce a yield strength dependence on temperature and strain rate, whereas long-range barriers are athermal in nature.

At low strain rates, below 10^4 s^{-1} , to overcome the resistance force of a short-range barrier, dislocations can use a combination of both thermal energy and an applied stress. The thermal energy, ΔG^* , can be defined as the total energy, ΔF^* , required to move a dislocation minus the mechanical work, W_M , done by load on a dislocation, Eq 1. The mechanical work can be broken down into an activation volume and applied stress, τ^* , where b is the burgers vector, l is the length of the dislocation and (x_2-x_1) is the distance the dislocation has to move.

$$\Delta G^* = \Delta F^* - W_M = \Delta F^* - \tau^* b l (x_2 - x_1) = \Delta F^* - \tau^* V^* \quad (1)$$

Taking the velocity of a dislocation (v_d) as the frequency of successful jump attempts, v , over a distance d , Eq. 2, and following Orowan's equation ^{22,23}, the strain rate, $\dot{\epsilon}$, can be written as Eq. 5:

$$v_d = v \cdot d \cdot \exp\left(\frac{-\Delta G^*}{kT}\right) \quad (2)$$

$$\epsilon = \rho_m b l \quad (3)$$

$$\frac{d\epsilon}{dt} = \dot{\epsilon} = \rho_m b \frac{dl}{dt} = \rho_m b v_d \quad (4)$$

$$\dot{\epsilon} = \rho_m \cdot b \cdot v \cdot d \exp\left(\frac{-\Delta G^*}{kT}\right) \quad (5)$$

There exists a critical temperature, T_c , where $\Delta G^* = \Delta F^*$ and the short-range barriers can be overcome by thermal energy alone. From this, it becomes clear that as the strain rate, $\dot{\epsilon}$, is increased, T_c increases due to less time for a successful jump from a thermal fluctuation. This work laid the foundational understanding that strength decreases with increasing temperature and increased with increasing strain rate, at rates below $\sim 10^4 \text{ s}^{-1}$.

However, as the deformation rate increase even higher, and approaches $\sim 10^5 \text{ s}^{-1}$, a more dramatic rise in strength is observed in almost all metals. This points to a change in deformation mechanism; however, the rate where it begins to occur also corresponds to the upper limit where most high-rate mechanical tests are intrinsically limited.

1.2 High-rate deformation mechanisms

As deformation rates approach and exceed strain rates of $\sim 10^4 \text{ s}^{-1}$, new deformation mechanisms begin to emerge. Dislocation motion transitions from thermally activated motion to ballistic transport through interactions with phonons and electrons in the crystal lattice. At these extreme strain rates, the engagement of rapidly moving dislocations with phonons becomes significant. What is more, the dislocations themselves are emitting phonons as they propagate which interact with the phonons in the crystal lattice. The phonons act as a source of drag on the dislocations, effectively pinning them, which inflates the total strength of a metal with increasing strain rate. Interestingly, however, theory predicts that as the temperature is increased from ambient, the phonon density increases. This can be quantified using the dislocation-phonon drag damping constant. Increases in this damping constant with increasing temperature indicate that, beyond some critical strain rate, the strength of a metal will increase with increasing

temperature. One of the first observations of the damping constant increasing with temperature by measuring dislocation motion is from Parameswaran et al.²⁴, who measured dislocation etch pits in Al single crystals impacted at different temperatures. From 30K up to room temperature, the damping constant increased with increasing temperature, seen as a decrease in dislocation velocity at a constant stress. While this trend is interesting on its own, there had not been a direct observation using a mechanical test of the strength of a metal increasing with temperature at very high strain rates.

Sargent and Ashby²⁵ were the first to include the dislocation drag regime in a deformation mechanism map by extending the maximum strain rates up to 10^6 s^{-1} for Ti, Fig. 2. In Fig. 2, contour lines within the “drag-controlled plasticity” region represent constant stress values. These contour lines are concave up, indicating that for a constant strain rate, as the temperature is increased the stress value also increases. In all other regions however, these stress contours are concave down meaning that as the temperature is increased the stress values decreases.

Sargent and Ashby defined the dislocation drag coefficient as a linear combination of two drag mechanisms, phonon drag and electron drag, Eq 6:

$$B = B_1 \frac{T}{300} + B_0 \quad (6)$$

where B is the overall drag coefficient, B_1 is the phonon drag contribution, B_0 is the electron drag contribution, and T is the temperature.

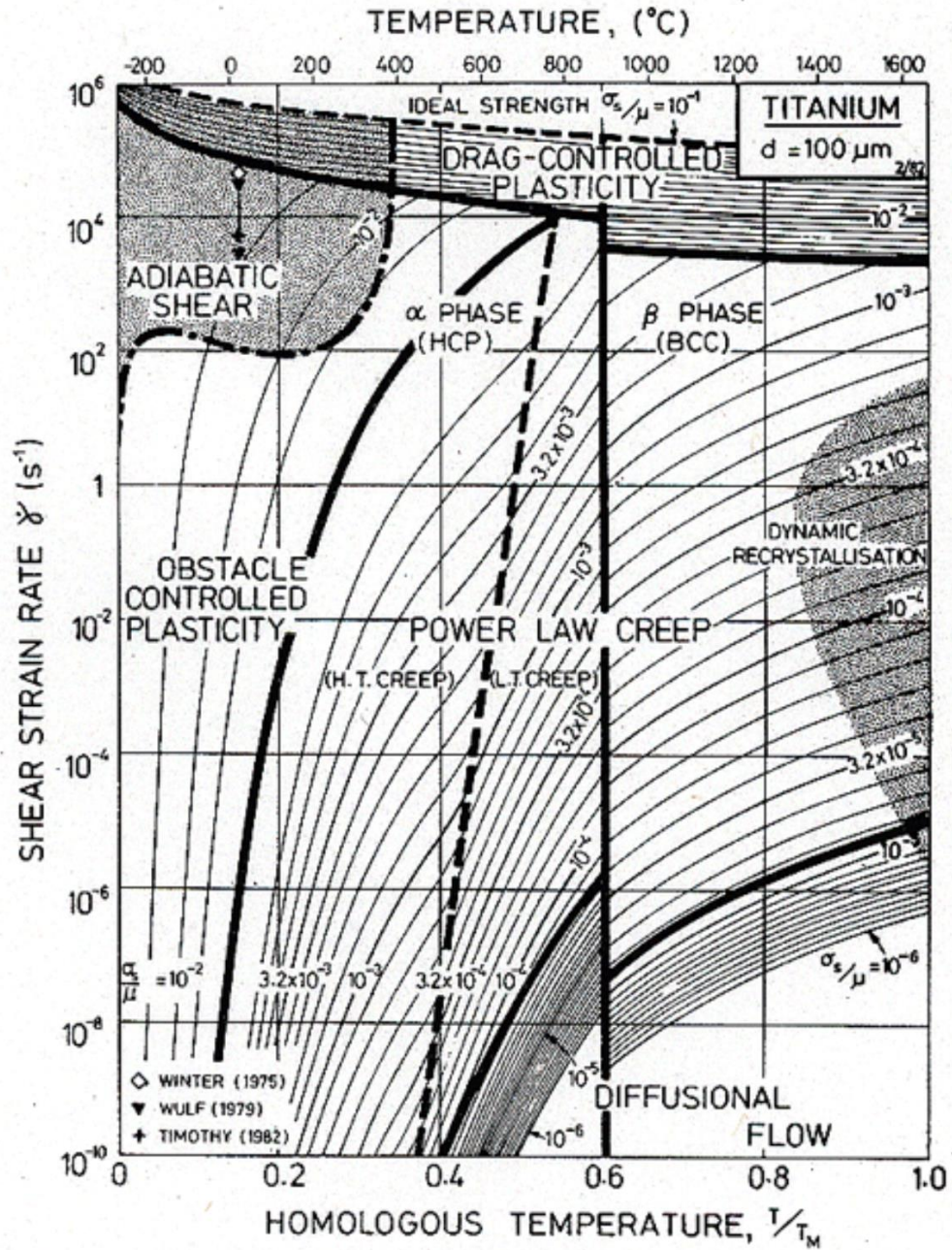


Figure 2: A high strain rate deformation mechanism map for Ti is shown, from Ref.²⁵. At the highest rates, above 10^4 s^{-1} , a drag-controlled plasticity region appears where the strength increases with increasing temperature.

While both of these terms are relevant at high strain rates, phonon drag effects only become significant above temperatures around 100K and electron drag effects are only significant at temperatures below 50K. This can be seen clearly in Fig. 3, where the electron and phonon drag coefficients, defined by Mason, are plotted as a function of temperature, dotted lines. Experimentally measured drag coefficients for a variety of metals are plotted alongside these theoretical curves, as show that as the temperature is increased, the electron drag mechanism “turns off” and the phonon drag mechanism “turns on”.

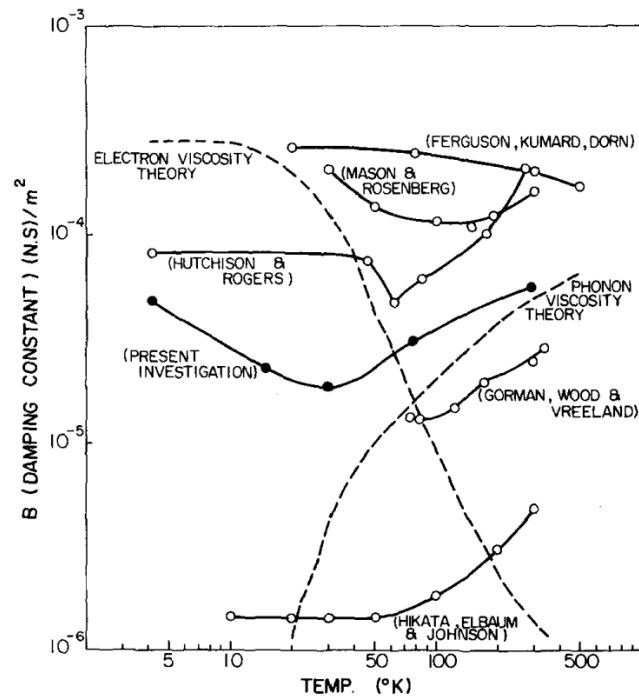


Figure 3: The dislocation damping constant for Al, solid black circles, is plotted along with the theoretical prediction for the electron and phonon damping coefficients from Mason, dotted lines²⁴. The damping coefficient seen experimentally is a linear combination of these.

Additionally, Sargent and Ashby noted the existence of an upper limit to the strain rate due to the relativistic contraction of the elastic field around the dislocation. At the time, strain

rates higher than 10^6 s^{-1} had not been reached experimentally so this was taken as the upper limit. It was also noted that, these high-rate deformation mechanisms were “poorly investigated, and the data for them are meager and uncertain.” Despite the theoretical interpretation of the phonon drag mechanism, there was a shortage of high-rate mechanical experiments to probe these regions of the deformation mechanism map.

Figure 4, from Sakino²⁰, shows the theoretical predictions of thermally controlled strengthening (labeled in the figure as Eq. 4), drag controlled strengthening (labeled in the figure as Eq. 5), and the combination of the two (labeled in the figure as Eq. 3). At a constant temperature, as the strain rate increases, the flow stress in the material increases.

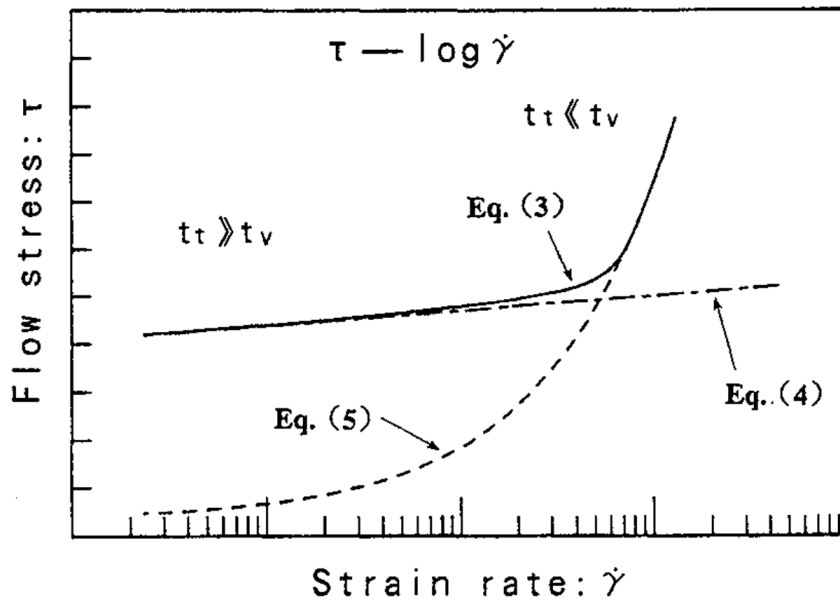


Figure 4: The theoretical prediction of the combination of thermal strengthening (labeled Eq. 4), drag strengthening (labeled Eq. 5), and the combination of the two (labeled Eq. 3). From Ref.²⁰

This prediction of an increase in strength with increasing strain rate, is consistent with the theory presented by Orowan²² above, and has been seen experimentally in many metals,

including by Armstrong²⁶ in copper, Fig. 5. While this data shows the beginning of the upturn in strength, due to dislocation damping, the data is limited to just where this transition starts and does not reach higher strain rates.

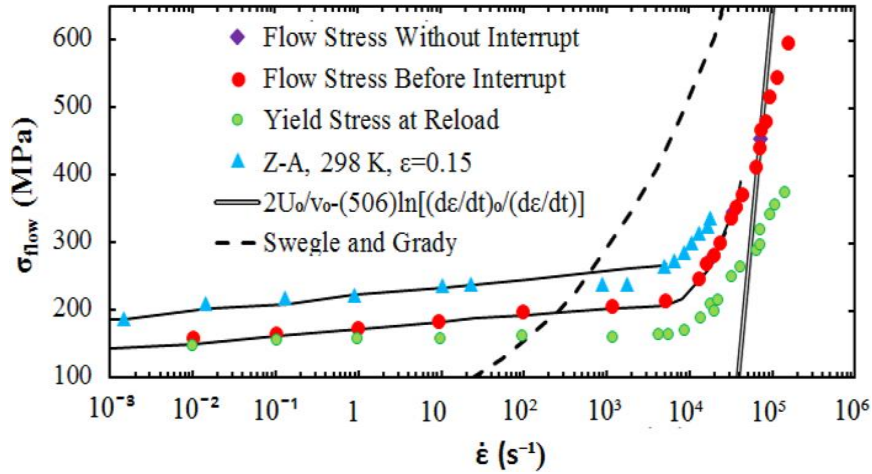


Figure 5: The flow stress in Cu as a function of strain rate is shown up to rates of $\sim 10^5 \text{ s}^{-1}$. Interestingly, a transition in the deformation mechanism is predicted due to the upturn in strength starting around 10^4 s^{-1} . From Ref.²⁶

1.3 Laser induced projectile impact testing

Advances in optically driven microballistics have made it possible to accelerate projectiles that are tens to hundreds of microns in size to velocities ranging from a few m/s to over 1 km/s. Most notably, the laser induced projectile impact test (LIPIT) has proved to be a useful experiment for studying material phenomena such as impact induced metallic bonding^{27–29}, erosion and hydrodynamic instabilities^{30–33}, impact induced melting^{34–37}, microscale additive manufacturing³⁸, and high strain rate mechanical properties^{39–41}. LIPIT experiments allow for one

microprojectile to be launched and viewed in-situ as it flies towards, impacts, and in some cases rebounds off of a target substrate material.

The standard set up and configuration of a LIPIT experiment, Fig. 6, is as follows²⁹. Launching pads used to accelerate projectiles are assembled with a 210- μm glass slide that has a 90-nm chromium layer sputtered onto the glass, and a uniform 30- μm polyurea film spun coat on top of the chromium layer, Fig. 8 b. Microprojectiles, commonly metallic or ceramic particles, are finely dispersed in a monolayer on top of a polymer film using a drop of ethanol and lens cleaning paper; upon drying the majority of the particles become isolated from each other on the surface. The launching pad is placed in a holder several hundred microns away from the target substrate material. With a long working distance microscope objective, the launch pad can be viewed from the side and individual microprojectiles can be selected for an impact experiment based on its size, shape, surface morphology, etc. Once a particle is selected, a 532-nm wave length laser pulse (Nd:YAG) of 10-ns duration is focused through the glass substrate onto the chromium layer, resulting in the ablation of the chromium.; this causes the polyurea to rapidly expand and the selected projectile to be launched towards the substrate. The velocity of the projectile is adjusted by tuning the energy of the laser pulse in a range of 1-80 mJ.

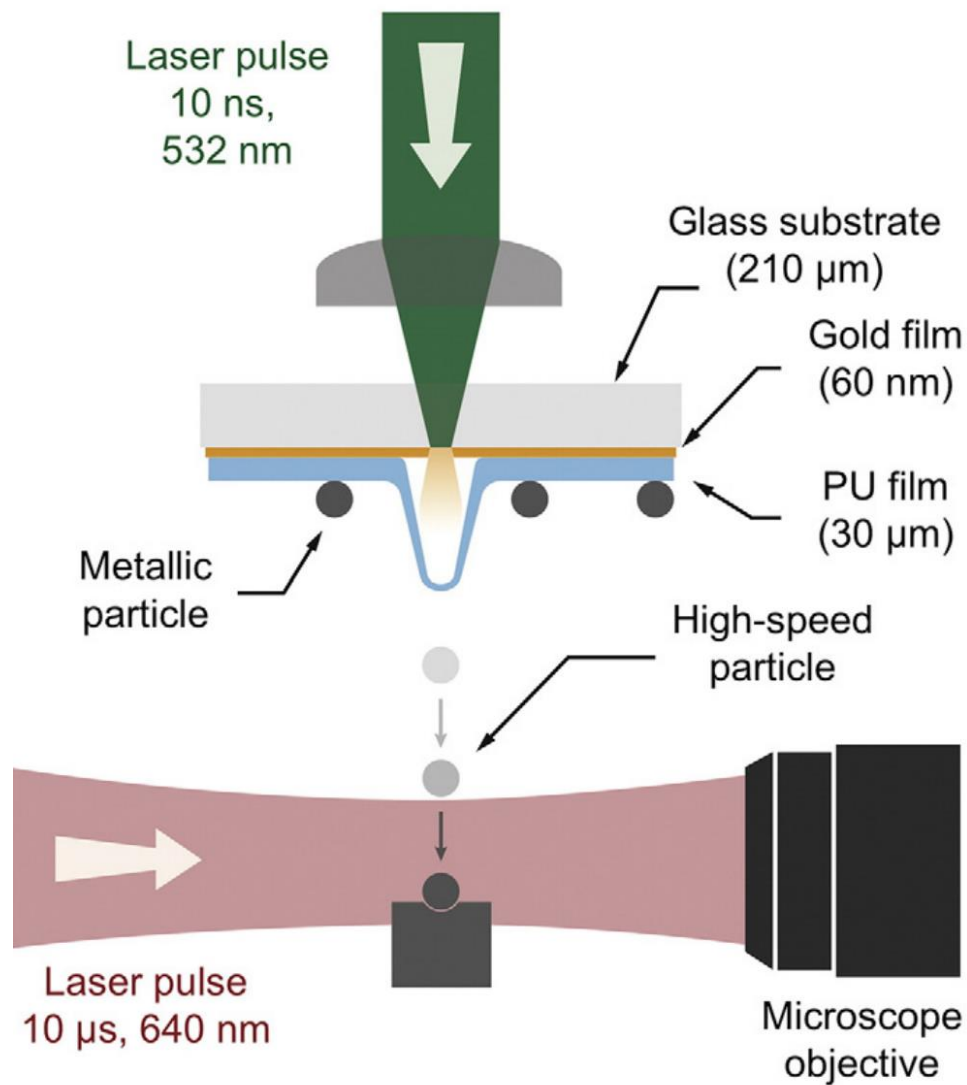


Figure 6: A schematic of the standard LIPIT set up is shown. Ref. ²⁹

The trajectory and impact of the projectile is captured using an ultra-high-speed camera with 16 independently triggered CCDs (Specialised Imaging), under the illumination of a 640-nm wavelength laser pulse, 10-μs duration, with 5-ns expose times, to produce a 16-frame sequence of the event. The image sequences provide direct measurements of inbound and rebound velocities and angles, as well as pre- and post- impact particle sizes. Figure 7, shows an image

sequence of a ceramic alumina particle in flight and rebounding off of a Cu substrate⁴⁰. The particle impacts the substrate at 425 m/s and rebounds at 40 m/s while its shape appears to remain unchanged before and after impact. Because of the high spatial and temporal resolution, the velocity measurements are accurate to within $\pm 2\%$.

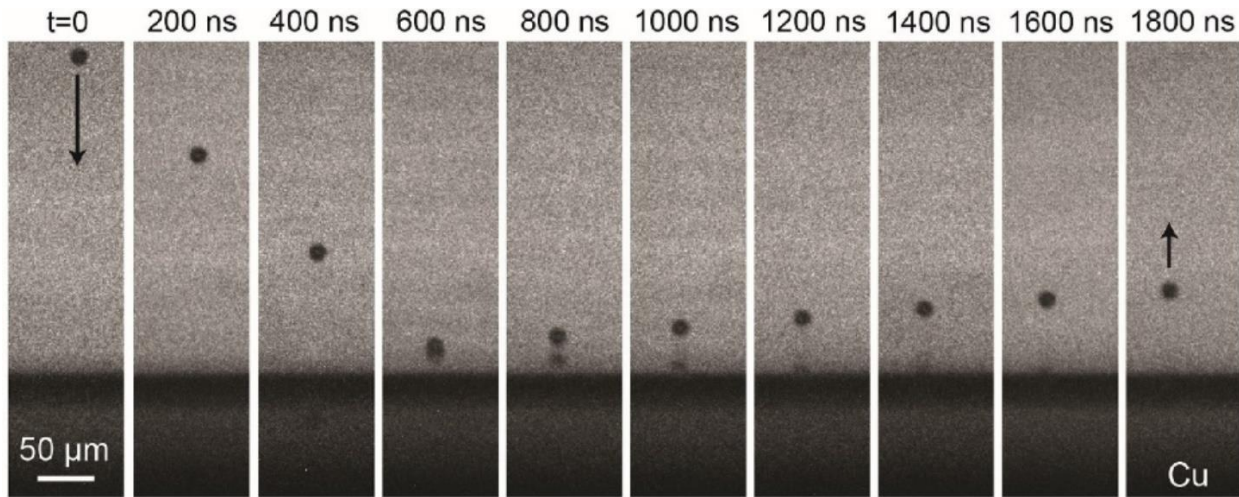


Figure 7: An exemplar image sequence of a 14- μm ceramic alumina microparticle impacting and rebounding off of a polished Cu substrate at 425 m/s and 40 m/s⁴⁰.

Because of the length scale of the projectiles and the velocities which they are launched, strain rates ranging from 10^6 – 10^9 s^{-1} are produced in the target material upon impact. This procedure includes particle-wise measurements of the impact velocity and particle diameter, thus providing a net strain rate for every individual experiment as v_i/d ; this quantity can be regarded as an average over the plastically deformed region over the duration of the impact event.

Despite their relatively high velocities, upon impact, the microparticles do not produce very high impact pressures. High impact pressure shock waves can propagate through the materials and convolute the mechanical response of the material. For impacts of steel particles on tin substrates, Lienhard et al.³⁰ found that even at the high impact velocity tested, 1020 m/s, the impact pressure was only 16.1 GPa. This upper bound of pressure at the shock front was estimated using Eq. 7:

$$P = \frac{1}{2} \rho v_s v_i \quad (7)$$

where P is the shock pressure, ρ is the density of the particle, v_i is the impact velocity of the particle, v_s is the velocity of the shock front in the substrate from the shock Hugoniot defined as:

$$v_s = c_o + \frac{1}{2} s v_i \quad (8)$$

where c_o is the bulk sound velocity of the substrate and s is a material specific constant.

This combination of low impact pressures and high strain rates makes microprojectile impact testing, here using LIPIT, an attractive approach for exploring materials under extreme strain rate deformations.

1.3.1 High temperature LIPIT

To extend microprojectile impact experiments to high temperatures, several modifications to the existing set up are needed. Launch pads containing a polymer layer are now insufficient to launch particles to any appreciable velocity due to the thermal softening and loss of elasticity of the polymer⁴². Above ambient temperatures, the polymer is replaced with a second glass slide, creating a sandwich that is a 210- μm glass slide with the 90-nm chromium layer sputtered onto it, a UV-curable adhesive and a second 210- μm glass slide without a metal layer sputter coated onto it, Fig. 8c. Using this set up, temperatures up to 300 °C can be tested without any material degradation.

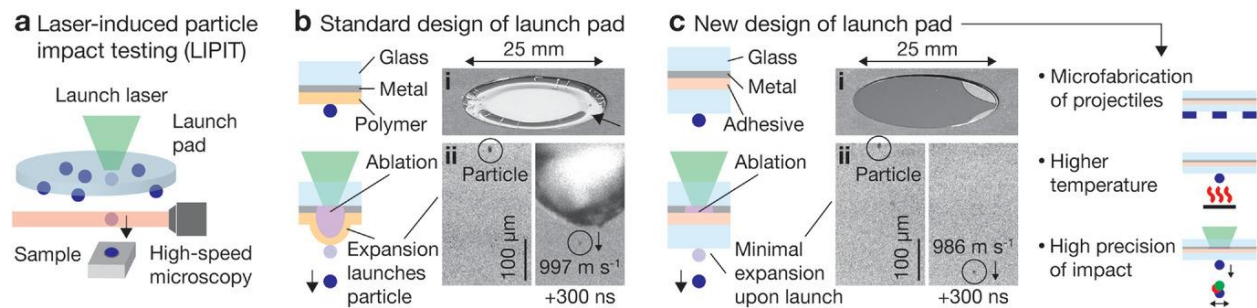


Figure 8: **a)** A schematic of the LIPIT with the glass launch pad assembly is shown. **b)** The standard launch pad design using the polymer layer is both schematically and experimentally shown. The expansion of the polymer bubble the accelerates the projectile forward is seen. **c)** The alternate glass sandwich launch pad design. Because the polymer layer is eliminated, there is no longer the expanding bubble or degradation of material when heated to several hundred degrees Celsius. Ref. ⁴²

To conduct impacts at elevated temperatures, a custom resistive heating stage was fabricated using a machinable ceramic (Macor, McMaster-Carr) and nichrome wire (McMaster-

Carr). Several thermocouples are placed around the top surface of the substrate to ensure that a homogenous temperature distribution was reached. Before launch, the projectiles are heated by virtue of their proximity with the target substrate, with an offset distance of only 750 μm ; the system was thermally equilibrated before launch, so particle, vapor and substrate are all at the same nominal temperature.

1.3.2 Using LIPIT as a mechanical testbed

One of the advantages of LIPIT is that it can be used to launch hard ceramic microparticles towards softer metal substrates. Under almost all impact velocities using LIPIT, a plastic indent is left in the surface of a metal substrate. The size of the indent increases as the amount of kinetic energy increases for each impact. When hard particles are used and remain undeformed before and after impact, as seen in the in-situ image sequences, these impact experiments can be used as a dynamic microindentation test. Because the size of the particle is known prior to launch, using the impact and rebound velocities gives a direct assessment of the amount of plastic work dissipated in the substrate for a given impact experiment.

Further, each impact crater can be evaluated individually using a complement of characterization techniques, most notably electron microscopy and laser scanning confocal microscopy. While electron microscopy gives qualitative information about things such as substrate melting, laser scanning confocal microscopy can be used to directly measure the volume of the impact crater. Hassani et al.⁴⁰ was the first to assess both the amount of plastic work dissipated in the substrate and the volume of the impact crater during impact experiments

using LIPIT. The ratio of plastic work to indentation volume gives an assessment of the dynamic hardness of a metal.

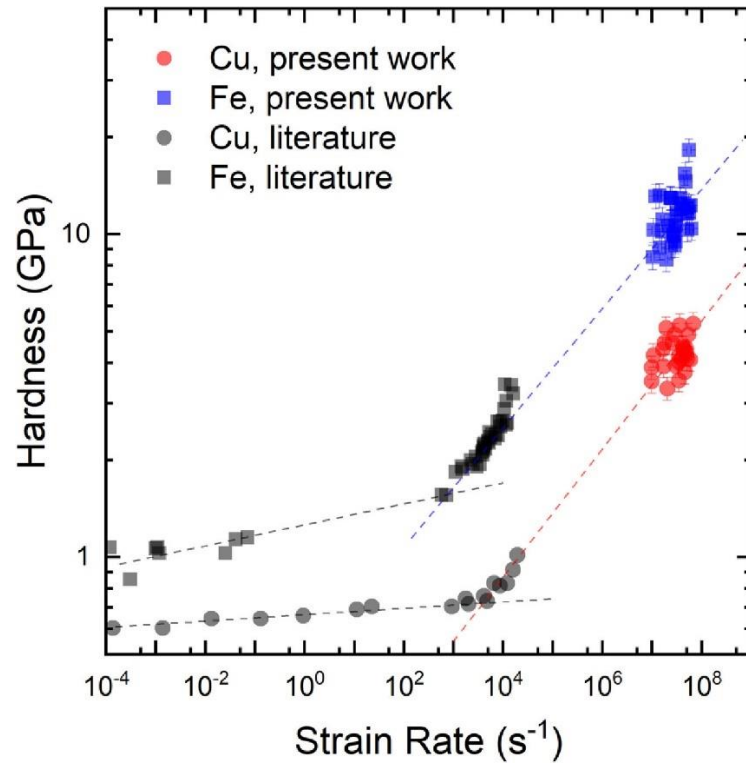


Figure 9: The dynamic hardness of copper and iron measured using single microparticle impact testing is shown as a function of strain rate. When compared to studies conducted at other strain rates, a dramatic increase in hardness is seen. Ref.⁴⁰

In Fig. 9, the dynamic hardness of Cu (red) and Fe (blue) was plotted as a function of strain rate. As the rate increased, the dynamic hardness for both materials increased. When connecting to lower rate literature data, it became clear that a new deformation mechanism region was reached, due to the dramatic increase in hardness seen, as well as a change in the scaling with strain rate.

Dynamic hardness measurements have also been performed on Al single crystals at three different orientations, by Cai et al. to study the effect of crystallographic orientation on hardness⁴¹. Under their impact conditions, there was no appreciable difference in the impact crater volumes for impacts at the same velocity on different orientations.

Beyond dynamic hardness measurements, the LIPIT offers the capability to test other high and extreme rate behaviors of materials. Because of the high precision of the LIPIT, erosion and ejection of material can be quantitatively studied. Hassani et al.³⁶, using matched material impacts (the same metal particle and substrate), provided an analysis that proposed the onset of erosion during a high strain rate particle impact, by balancing the input kinetic energy with the thermal properties of the materials:

$$v_{i,e} = \left(\frac{20 \cdot (\rho k C_p)^{1/2} \cdot \left(T_m - T_0 \frac{H_f}{C_p} \right)}{\rho_p \sqrt{d}} \right)^{2/5} \quad (9)$$

where $v_{i,e}$ is the onset erosion velocity, ρ is the density of the substrate, ρ_p is the density of the particle, d is the particle diameter, T_0 is the initial temperature, T_m is the melting temperature, C_p is the specific heat, k is the thermal conductivity, and H_f is the heat of fusion. This approach is valid for any particle-substrate pairings under ambient conditions and is not limited to only matched materials. The first term in the numerator is the thermal effusivity, e_{th} , and the second

term in the numerator is the melting index, I_{melt} . Figure 10. serves as a map for the onset of melting for any set of materials.

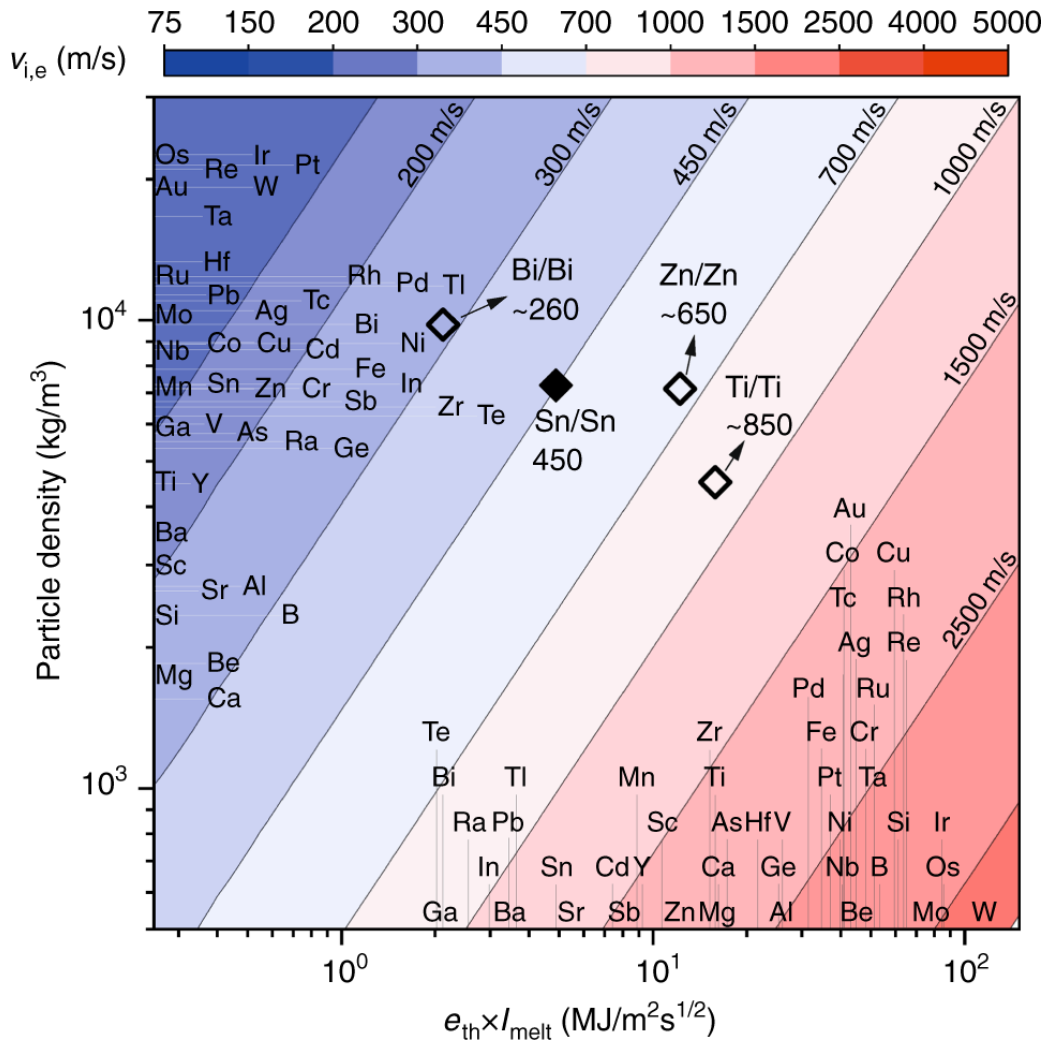


Figure 10: The velocity for the onset of erosion can be predicted for any particle substrate pairing from Eq. 9³⁶.

Lienhard et al. conducted a similar set of experiments, however used hard steel particles on soft tin substrates to isolate the deformation to the substrate³⁰. Because the high precision of

the LIPIT allows for each experiment to be studied post impact, scanning electron micrographs of each impact crater can be used to observe the onset of melting and erosion. By systematically increasing the impact velocity in a small step size, the onset of melting was experimentally identified for this system was observed to be 310 m/s. Figure 11, from Ref.³⁰ shows the progression of plasticity and onset of melting as the impact velocity is increased. The onset of melting can clearly be seen in the impact at 310 m/s with the degree of melting increasing as the impact velocity increased. Turning to Fig. 10 from Hassani et al., this experimentally assessed value for the onset of melting is in agreement with Eq. 9, which predicts the onset of melting to occur at ~ 300 m/s.

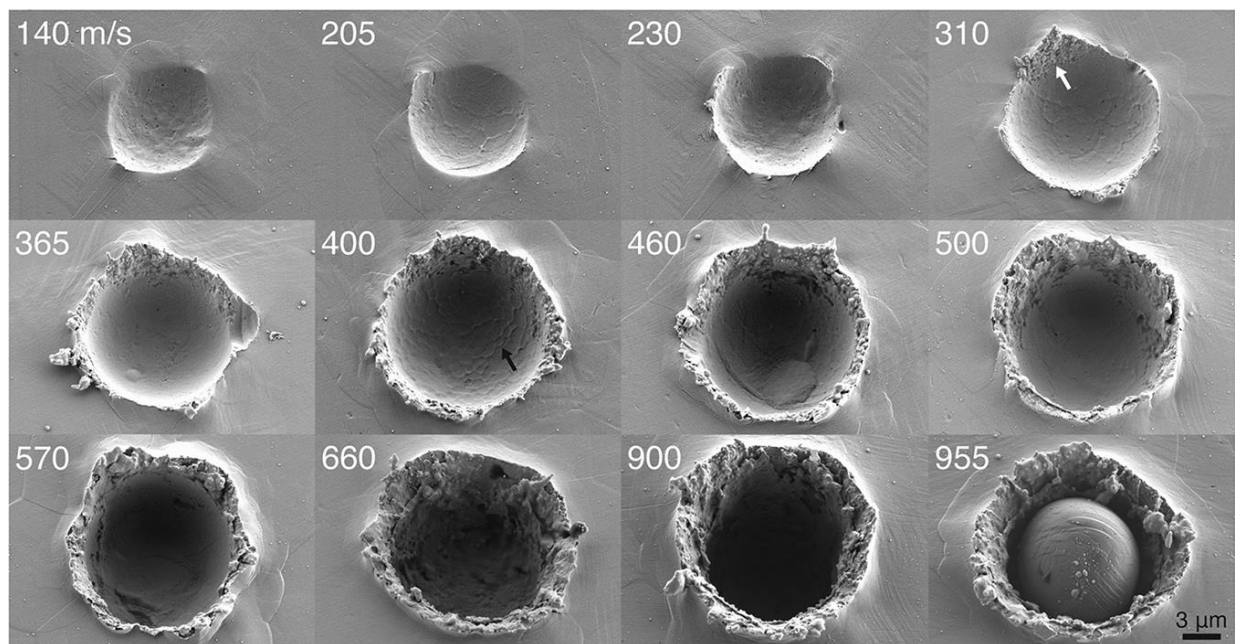


Figure 11: The impact craters from high velocity impacts of steel particles on to tin substrates are shown for increasing impact velocities. At 310 m/s the first signatures of melting are seen. From Ref.³⁰.

The LIPIT proves to be a valuable tool that complements others high strain rate experiments. Due to its high degree of adaptability, the LIPIT provides insights into many impactor/substrate material pairs where severe plastic deformation processes can be challenging to study in a systematic way. Combining the LIPIT with other characterization techniques, allows it to become a new quantitative tool for studying material properties at strain rates $>10^6 \text{ s}^{-1}$.

1.4 Thesis Outline

This thesis quantifies and develops an understanding of the extreme rate mechanical properties of metals. Using microprojectile impact experiments to produce deformation strain rates above 10^6 s^{-1} and a complement of post impact characterization techniques, the dynamic mechanical properties of metals can be assessed at ambient and elevated temperatures. Trends in dynamic strength and dynamic hardness under these conditions are explored and the boundaries of where certain trends happen is investigated.

Chapter 2: The dynamic mechanical properties of pure metals is first investigated using microparticle impact experiments of ceramic alumina particles on pure metal substrates at three different temperatures. These experiments provided the first look into how the dynamic hardness changes with temperature in pure Cu, Ti, and Au. A strength model describing this behavior was developed and shown that it not only matches the experimental data, but that the observed trends are the thermodynamically and kinetically expected forms of plasticity in metals.

Chapter 3: The work from Chapter 2 is then extended to metals containing solute elements. The addition of solute elements now provide a new strengthening mechanism for

dislocations but increasing the number of pinning points. Solutes are systematically added to pure Ni at ambient and elevated temperatures to probe the phonon drag dominated plasticity regions as a function of solute content. A new strength model to describe this behavior is introduced.

Chapter 4: Any strengthening mechanism in a metal has an upper limit temperature where it breaks down. This temperature is generally at the melting temperature where the strength of a metal effectively goes to zero. The properties of low melting temperature Sn are assessed by investigating how heating up close to the melting temperature effects the rebound behavior and dynamic hardness when adiabatic heating from a high-rate impact begins to melt material.

Chapter 2: Pure metals increase strength with increasing temperature at extreme strain rates

Metals usually soften at high temperatures, because the motion of defects that control strength—dislocations—is thermally activated: temperature helps these carriers of plasticity move across the crystal lattice and bypass any obstacles there^{9,43–45}. The softening of metals with temperature is extremely broadly observed, being the norm over more than 10 orders of magnitude in strain rates, spanning from 10^{-6} s^{-1} – 10^4 s^{-1} . However, as strain rates increase beyond 10^4 s^{-1} , additional deformation mechanisms can become active; these contribute to a significant increase in strength through dislocation drag controlled plasticity^{19,25,46,47}. Understanding material properties at such high strain rates is critical for designing and engineering new materials for use in extreme conditions. Deformation rates above 10^4 s^{-1} are commonly seen in hypervelocity meteorite impacts⁴⁸, high speed metal machining⁴⁹, sandblasting and erosion^{36,50}, and in metal additive manufacturing processes such as cold spray¹. The drag mechanisms that prevail under high rates are interesting because they are not rate limited in the same manner as the thermally activated mechanisms described above; they are instead expected to be a mode of *ballistic* dislocation transport. It has been anticipated that in this extreme range, metals indeed may not soften at high temperatures^{20,25,47,51}, a behavior that is counterintuitive and at odds with decades of studies in less extreme conditions.

Unfortunately, experimental data on extreme strain-rate deformation mechanisms are rather rare, due to limitations of measurement tools that can access extreme strain rates. Kolsky bars⁵², flyer plates⁵³, shock impacts⁵⁴, high-power laser pulses⁵⁵, shockless isentropic compression experiments⁵⁶, and gas gun driven projectiles⁵⁷ are examples of methods that are

generally conducted on macroscale samples (ranging from mm to a few hundred mm) and require high impact velocities (hundreds of meters per second) to induce these large strain rates. In general, the quantitative measurements provided by these methods end around a strain rate of $\sim 10^4 \text{ s}^{-1}$. At higher rates these approaches are more typically aimed at studying material spall⁵⁸, and convolute the mechanical response with strong shock effects in the material during impact. We are not aware of data that are able to separate out shock effects and show the posited loss of thermally-activated softening at high strain rates.

Advances in optically driven microballistic systems over the last decade have made it possible to accelerate $\sim 1\text{-}50 \text{ }\mu\text{m}$ sized impactors to velocities up to and beyond 1000 m/s, producing extreme strain rates from $10^6 - 10^9 \text{ s}^{-1}$, strain rates on the same order of magnitude as, e.g., discrete dislocation and molecular dynamics simulations^{42,59}. Importantly, due to the small scale of the test, these rates are achieved without necessarily passing into the strong shock regime that characterizes many other rapid experimental test methods, permitting a clean assessment of material strength under extreme rates. Even more interestingly, there are now at least two different means of using such tests to extract quantitative measures of strength from an impacted substrate: dynamic yield strengths can be extracted from the rebound behavior of the impactor over a range of velocities⁶⁰⁻⁶², and dynamic hardness values can be measured based on post-impact measurements of the impact crater sizes⁴⁰. The availability of these strength measures at extreme rates opens the door to exploration of the drag mechanisms that are expected to dominate there, but are quantitatively underexplored.

2.1: Methods

Spherical alumina powder particles, with nominal particle size of $\sim 12.5 \pm 1 \mu\text{m}$, were purchased from Inframat Advanced Materials (Manchester, CT) and used for microparticle impact experiments on copper and titanium, and nominal particle size of $\sim 20 \pm 1 \mu\text{m}$ for impact experiments on gold. Oxygen free high conductivity copper plates (OnlineMetals, USA) and pure Ti rods (Beantown Chemical, USA) were cut using a diamond saw and annealed at 600°C for 10 hours. Pure gold rods (Beantown Chemical, USA) were cold rolled to an 85% reduction in thickness before being annealed at 125°C for 3 hours. The plates and rods were then polished using standard metallographic techniques to a $0.04 \mu\text{m}$ mirror finish.

The velocities targeted in this study (below 250 m/s for copper and titanium and below 150 m/s for gold) are all quite low compared with conventional high-rate testing; this is far below the point where instabilities associated with hydrodynamic jetting/spall set on (~ 650 for the present geometry in Ref. ⁶²), and even farther from the regime of hydrodynamic penetration (~ 900 from Ref.³³). Every impact falls within a regime where no fracture-like events occur that could lead to fragmentation and ejection of the substrate material. The onset of such events would be identifiable by an excess loss of rebound velocity, seen as deviations from the power-law in Fig 13a⁶², as well as post-mortem analysis of the craters³⁰; nothing seen in our data suggests any material fragmentation or ejection, as expected for these conditions. Additionally, all of the impact velocities for each material are also well above the elastic-plastic transition velocity (v_y from Eq. 11). Plastic deformation sets on at velocities below 1 m/s for each material in this study, meaning that each impact in this work leaves a plastic crater in the substrate. Our procedure includes particle-wise measurements of impact velocity and particle diameter, thus

providing a net strain rate for every individual experiment as v_i/d ⁴⁰; this quantity can be regarded as an average over the plastically deformed region over the duration of the impact event.

2.2: Quantitative measurements of strength and hardness

High strain rate micromechanical impact experiments were carried out using laser induced particle impact tests, detailed in the Chapter 1. Spherical alumina microparticles, $\sim 12.5 \pm 1 \mu\text{m}$ in size, were chosen as impactors onto copper substrates due to their much higher hardness (static hardness 15.7 vs. 0.56 GPa for annealed Cu)^{63,64}, to isolate the plasticity to the substrate. Figure 12 shows some typical results, in which we track by high-speed camera the inbound and rebound trajectories of alumina particles as they impact the copper substrate. The three examples in Fig. 12 are chosen for the close similarity of the conditions: these three nominally identical particles are launched at almost the same velocity, and thus follow the same inbound trajectories very closely. However, these three experiments were conducted at 20, 100, and 177 °C, and as the test temperature is increased, the particle rebounds more aggressively, as shown by the increasing rebound trajectories as well as the differences in height in the inset black and white photographs of the particles 600 ns after impact. These results speak to a higher strength of copper at higher temperatures, as more kinetic energy is carried away in the rebound; this is a first signature of anomalous temperature dependence of strength at high rates.

The raw trajectory data in Fig. 12 are corroborated by an independent measurement of local plastic deformation. The upper insets in Fig. 12 show impact crater profile measurements using 3D laser scanning confocal microscopy for the same three impacts, in profile (top row) and in plan view projection (middle row); the crater diameter and depth decrease with test

temperature at the same impact velocity. Quantitatively, the crater volume drops by about a factor of two between room temperature and 177 °C. Since crater volume is inversely proportional to strength ⁶⁵, the crater images in Fig. 12 independently confirm that copper is hardening at higher temperatures under these extreme impact conditions.

The two independent signatures of thermal hardening in Fig. 12—the increased rebound and the decreased crater volume—are both amenable to quantitative analysis to extract strength measures, and in what follows we take each of these in turn. We first look at a larger set of rebound data for many more impact experiments. The rebound response is quantified using the coefficient of restitution, $CoR = v_r/v_i$, the ratio of the rebound and impact velocities. Figure 13a plots the CoR as a function of the impact velocity in double logarithmic format. For a given test temperature, higher impact velocities lead to lesser rebounds, as more energy is dissipated in plasticity of the substrate. In fact, the parabolic scaling in Fig. 13a is the expected form for plasticity, as developed by Wu et. al. ^{60,61}, who showed that ideally plastic impacts follow the power-law relation:

$$CoR = \frac{v_r}{v_i} = \alpha \left(\frac{v_i}{v_y} \cdot \frac{E^*}{Y_d} \right)^{1/2} \quad (10)$$

where the coefficient $\alpha = 0.78$ for an undeforming impactor on a deforming substrate, Y_d is the dynamic yield strength of the substrate, E^* is an effective elastic modulus of the materials involved, and v_y is a characteristic velocity at which plastic deformation is initiated:

$$v_y = \left(\frac{26Y_d^5}{\rho E^{*4}} \right)^{1/2} \quad (11)$$

where ρ is the density of the substrate.

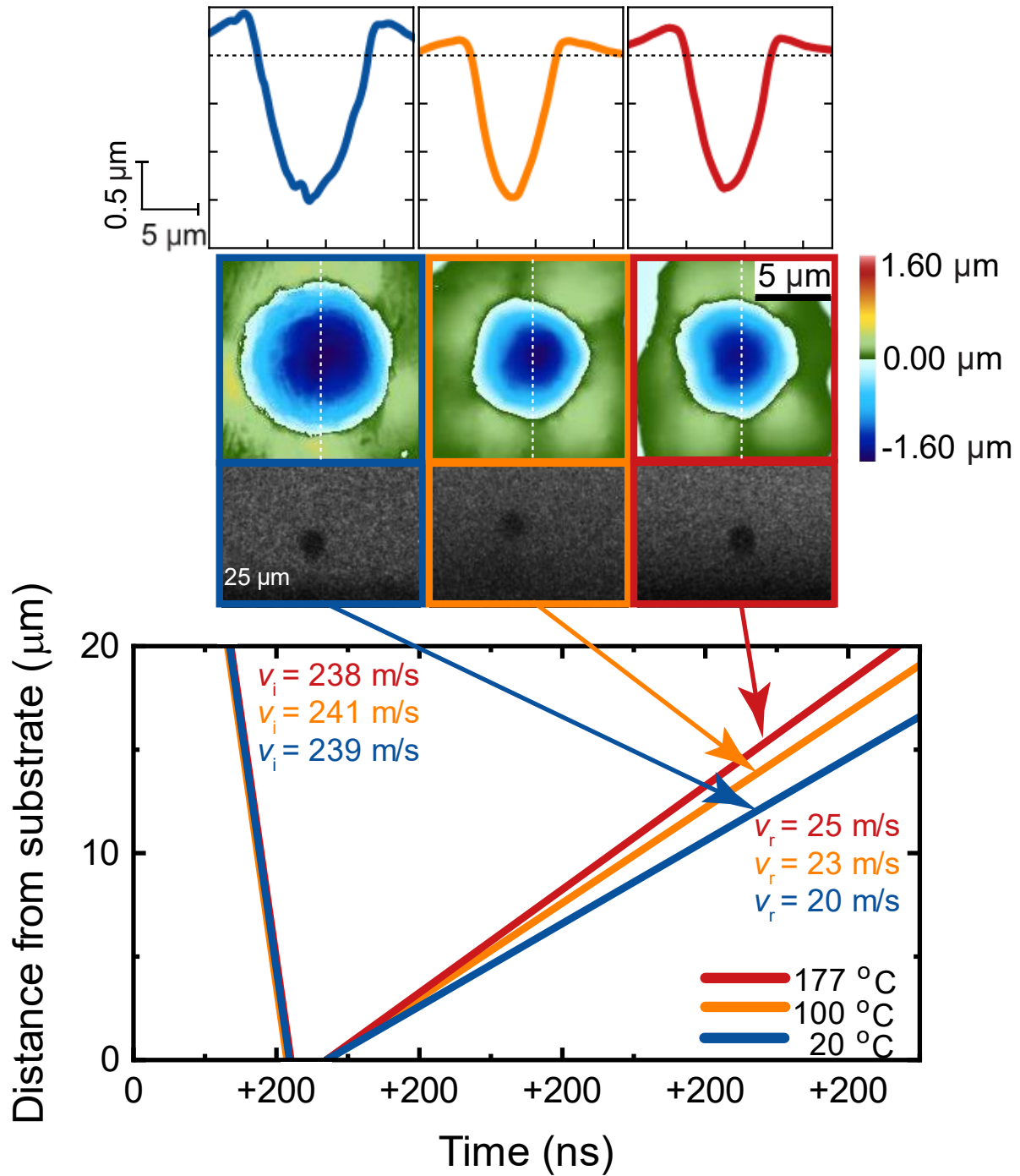


Figure 12: The impact and rebound trajectories of three alumina particles impacting a copper substrate at 20 $^{\circ}\text{C}$ (blue), 100 $^{\circ}\text{C}$ (orange), and 177 $^{\circ}\text{C}$ (red). As the temperature is increased, the rebound velocity of the impacting particle increases, as shown by both the plotted rebound trajectories and gray insets of the particle 600 ns after rebound. Insets of plan and line scan views of each impact crater, measured with 3D laser scanning microscopy, show that as the temperature is increased the impact crater depth and width decrease.

Although microballistic experiments are not elastic-perfectly plastic conditions, fitting with this model returns an average measure of Y_d over a large range of impact velocities averaging strain- and rate-hardening, adiabatic heating, grain size evolution, etc., into an effective strength of the impact event.

Not only do the data in Fig. 13a conform to the scaling of Eq. (10), the equation is fully predictive with the fitting of a single parameter, namely the dynamic yield strength Y_d , which is an average strength measure describing the complex deformation of the substrate over the entire duration of an impact. At room temperature, fitting the CoR data in Fig. 13a yields a dynamic strength of 280 MPa for copper at the average strain rate of 10^7 s^{-1} for these tests. For comparison, under quasi-static conditions annealed copper would have a typical strength on the order of 70 MPa⁶⁶, and under extreme conditions involving shock (such as Richtmeyer-Meshkov instability tests⁶⁷) strengths up to 1000 MPa have been inferred. The present value between these limits represents significant rate-hardening, but no superimposed effect of shock or high pressure. One advantage of this assessment of strength is that it averages out local heterogeneities in the microstructure (including grain boundaries, crystal orientation effects, defect structures) over the entire range of impact velocities; these effects contribute to the scatter in the data, and provide a quantitative measurement uncertainty as a standard deviation of the data's departure from the trendline in Fig. 13a.

The more interesting trend in Fig. 13a is the effect of the test temperature on the CoR curves and the dynamic yield strengths they encapsulate. Just as seen before in Fig. 12 for a few example cases, we now see very broadly that temperature promotes greater rebounds in copper, over the full range of tested velocities. The fitted solid lines reflect an increasing dynamic

strength: $Y_{d, 20^{\circ}\text{C}} = 280 \text{ MPa}$, $Y_{d, 100^{\circ}\text{C}} = 310 \text{ MPa}$, and $Y_{d, 177^{\circ}\text{C}} = 364 \text{ MPa}$. With increasing temperature, the dynamic yield strength of copper increases by perhaps 30% over a range of $\sim 150^{\circ}\text{C}$.

One limitation of the CoR analysis above is that it averages the dynamic strength over a range of velocities because it requires fitting of many data points together. A second strength measure, the dynamic hardness, permits a test-by-test analysis of the trends revealed above. Dynamic hardness measurements also require the quantitative measurement of inbound and outbound velocities to assess the amount of energy deposited into the substrate and expended as plastic deformation, normalized by the crater volume, V ⁴⁰:

$$H_d = \frac{1/2 \cdot m_p \cdot (v_i^2 - v_r^2)}{V} \quad (12)$$

where m_p is the mass of the particle. The crater volume can be accurately assessed with scanning confocal microscopy as in Fig. 12, and combined with the associated velocity measurements gives the dynamic hardness of copper for each individual experiment. The results of this analysis are plotted in Fig. 13b for the range of strain rates tested, with the net strain rate of a given experiment assigned as the ratio of the impact velocity to the particle diameter, (v_i/d) . Convincingly beyond the intrinsic scatter of the experiments, these data confirm all of the observations above: at strain rates above about 10^6 s^{-1} , the hardness of copper increases with increasing test temperature.

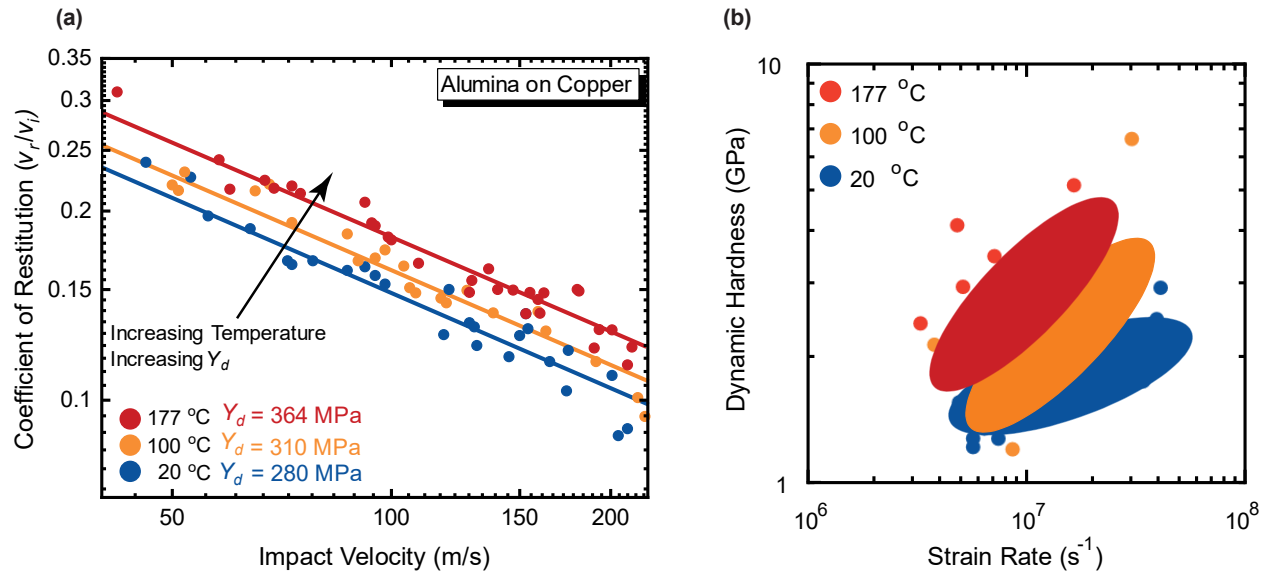


Figure 13: **(a)** The impact velocity versus coefficient of restitution, v_r/v_i , is plotted on a double logarithmic scale for a range of impact velocities. Across the span of impact velocities, the impacts at higher temperatures led to a faster rebound. The solid lines represent the scaling law for ideally plastic impacts with the fitting parameter, Y_d , increasing with temperature. Uncertainty in measured velocities is $\pm 2\%$, which is on the order of the size of the data markers. **(b)** Dynamic hardness measurements from each impact are plotted against deformation strain rate. The shaded regions serve as a guide for the eye to show where the majority of the hardness values fall for each temperature. As with strength, the hardness trends upward with increasing temperature, and measurement uncertainties are on the order of the size of the data markers.

The above results represent the first experimental data to show anomalous temperature dependence at strain rates above $10^6 s^{-1}$, in experiments that are both quantitative and free of strong shock effects. This result is also not limited to pure copper; we have observed the same trends in pure Ti and Au as well.

Au was selected because it is a face-centered cubic metal like copper, but is noble and therefore doesn't have a native oxide layer and is thus likely chemically cleaner than copper. Second, titanium was selected to generalize the observation to a noncubic (hexagonal) crystal system.

Figure 14 shows the impact velocity versus coefficient of restitution (CoR) curves for alumina impacts on pure gold at 20 °C (blue) and 97 °C (red). The solid blue and red lines show the same parabolic scaling that is seen in copper, and can therefore be used to extract the dynamic yield strength, Y_d , by fitting with Eq. 10. A 77 °C increase in temperature led to ~5% increase in Y_d ; fitted values are shown inset to Fig. 14. Additionally, in Figure 14b, the dynamic hardness of titanium was measured for each impact at each temperature. In general, for impacts at the same strain rates, higher hardness values are measured for impacts at 97°C.

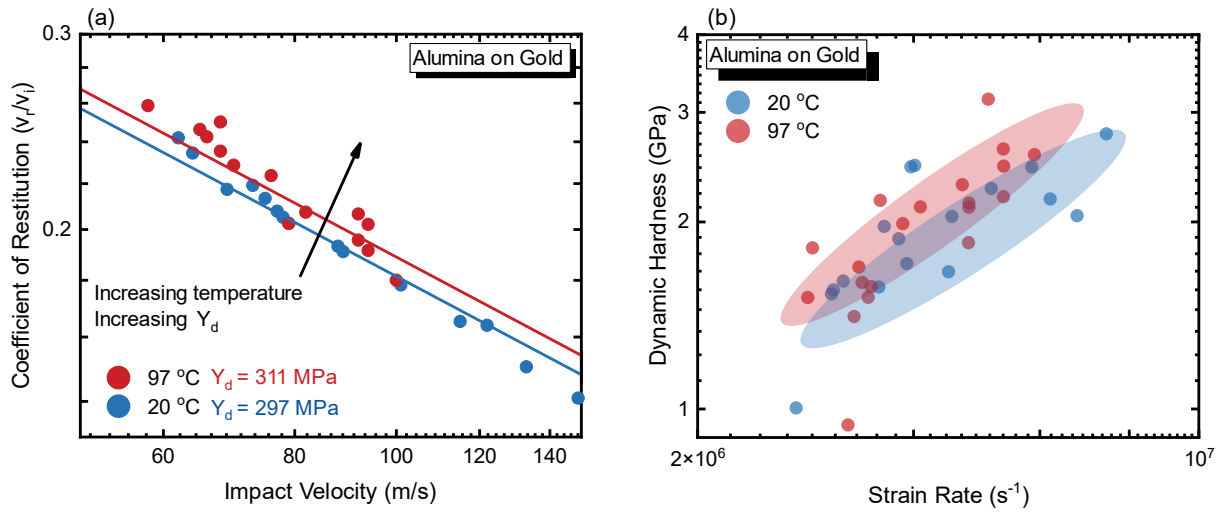


Figure 14: **(a)** The impact velocity versus CoR is plotted on a double logarithmic scale for alumina impacts on gold substrates at 20 °C (blue) and 97 °C (red). Using scaling for ideally plastic impacts the dynamic yield strength for each impact temperature is calculated to be $Y_{d, 20^\circ C} = 297$ MPa and $Y_{d, 97^\circ C} = 311$ MPa. Uncertainty in measured velocities is $\pm 2\%$, which is on the order of the size of the data markers. **(b)** The dynamic hardness of titanium is plotted for both 20 °C (blue) and 97 °C (red). For impact at similar strain rates, the tests at 97 °C show a higher hardness, and measurement uncertainties are on the order of the size of the data markers.

As with both copper and gold, the same trend is seen in titanium where a 180 °C increase in temperature led to a ~14% increase in Y_d , Figure 15a. What is more, for impacts at the same

strain rate hardness values at 200 °C were on average higher than hardness values at 20 °C, Figure 15b.

For all three metals, the two independent strength measures (dynamic yield strength and dynamic hardness) reflect a change in deformation mechanism due to the synergy between strain rate and temperature; at extreme strain rates, all three metals harden when heated.

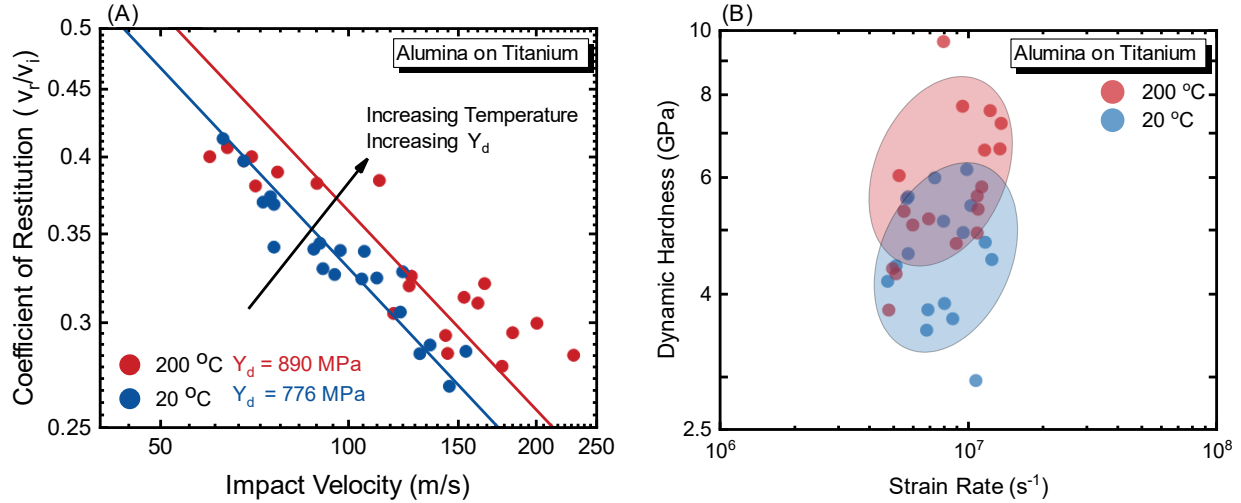


Figure 15: (a) The impact velocity versus CoR is plotted on a double logarithmic scale for alumina impacts on titanium substrates at 20 °C (blue) and 200 °C (red). Y_d for each impact temperature is calculated to be $Y_{d, 20^\circ C} = 776$ MPa and $Y_{d, 200^\circ C} = 890$ MPa. Uncertainty in measured velocities is $\pm 2\%$, which is on the order of the size of the data markers. (b) H_d of titanium is plotted for both 20 °C (blue) and 200 °C (red). For impact at similar strain rates, the tests at 200 °C show a higher hardness, and measurement uncertainties are on the order of the size of the data markers.

2.3: Deformation Mechanisms

The flow strength in pure metals can be broken down into three additive components: thermal, athermal, and drag strengthening^{68–70}, each of which is plotted separately in Fig. 16 at our experimental strain rate of $10^7 s^{-1}$:

- The thermal strength component, σ_{th} , comes from dislocation interactions with short-range barriers (Peierls-Nabarro barriers, vacancies), that can be fully overcome by thermal fluctuations. This strength can be calculated as a function of temperature and strain rate following Refs. ^{71–73} and follows a conventional thermally-activated softening-with-temperature trend at all rates and temperatures in metals (red line in Fig. 16).

$$\sigma_{th} = \left[1 - \left(\frac{kT}{g_0 \mu_T b^3} \ln \left(\frac{\dot{\epsilon}_0}{\dot{\epsilon}} \right) \right)^{1/q} \right]^{1/p} \sigma_0 \quad (13)$$

$$\mu_T = \mu_0 - \sqrt{a + \left(\frac{T}{T_r} \right)^2} \quad (14)$$

The thermal strength is calculated with Eq. 13, where σ_0 is the stress to overcome the relevant short range barrier at 0 K, k is the Boltzmann's constant, T is the temperature, g_0 is the activation energy normalized to 0 K, μ_T is the temperature dependent shear modulus (Eq. 14), b is the magnitude of the Burgers vector, $\dot{\epsilon}_0$ is a reference strain rate, $\dot{\epsilon}$ is the strain rate, and q and p are fitting parameters based on the shape of the short-range barrier profile ²¹. For copper, the temperature dependent shear modulus, Eq. 14, μ_0 is the shear modulus at 0 K, 47.1 GPa, and a and T_r are constants ⁷³ equal to 0.14 and 60.16, respectively.

- The athermal strength component, σ_a , arises from other barriers to dislocation motion that are not overcomable by thermal fluctuations alone, such as dislocation–grain boundary and dislocation–forest interactions ⁷³.

$$\sigma_a = \frac{\alpha_G \mu_T \sqrt{b}}{\sqrt{D}} + \alpha_{disl} \mu_T b \sqrt{\rho} \quad (15)$$

Dislocation-grain boundary strengthening effects can be captured in a Hall-Petch type relationship, the first term in Eq. 15. D is taken to be the final grain size after impact, on the order of several hundred nanometers based on copper microparticle impacts of Tamiyu et. al ⁷⁴, and α_G is a dislocation-grain boundary interaction parameter, here 0.15. The second term in Eq. 15 relates dislocation-forest interactions to strength through the mobile dislocation density, ρ ($\sim 10^{13} \text{ m}^{-2}$ at these strain rates in copper ⁷⁵) and a dislocation-dislocation interaction parameter, α_{disl} , equal to 0.5. Despite there being a small decrease in athermal strength with temperature due to a reduction in the temperature dependent shear modulus, these long-range obstacles cannot be overcome through increases in temperature.

The total athermal strength component of copper can be calculated with standard models for our strain rate of 10^7 s^{-1} , giving the blue line in Fig. 16, again a softening-with-temperature trend. The softening in this case is due to the lowering of the elastic modulus with temperature, which in turn lowers the energy of dislocation-obstacle interactions.

- The final strength component in pure metals, drag strengthening, comes from the interaction of dislocations with phonons in the lattice ⁷⁶. This interaction gives rise to anomalous thermal strengthening, because higher temperatures increase the energy levels of phonons and thus increase the interaction of dislocations with them.

Several sources of drag exist on moving dislocations in crystals, however under the temperature and strain rate conditions in this study, the dominant drag strengthening mechanisms are from interactions with phonons, namely phonon scattering and phonon viscosity.

$$\sigma_d = \frac{B_d}{\rho b^2} \dot{\epsilon} \quad (16)$$

$$B_d = \frac{3kT}{5 \cdot C a^3} (1.8b) \quad (17)$$

Equations 16 and 17 are a standard model for the phonon drag strengthening mechanism ⁷⁷, where B_d is the damping constant. In copper at strain rates of 10^7 and temperatures between 20 – 177 °C, B_d is $\sim 10^{-5}$ Pa·sec, and is consistent with both shock wave loading ⁷⁵ and atomistic simulations of copper at high rates ⁷⁸. The damping constant can be calculated as the effective dislocation width, $1.8b$, times the thermal energy density divided by five times the shear wave velocity, C , Eq. 17. The damping coefficient scales linearly with temperature and the lattice coefficient, a , marginally increases with temperature through the coefficient of thermal expansion. At a constant high rate, as in

Fig. 16, when evaluating σ_d as a function of temperature, the mobile dislocation density, ρ , is a constant ⁷⁵, and the strength changes only through the change in temperature and lattice parameter. Increases in T dominate any increases in a giving rise to the increase in strength across all temperatures at high strain rates. At sufficiently low temperatures, below around ~100K, phonon drag effects become negligible. While other sources of drag strengthening are possible ⁷⁹, such as electron drag ⁸⁰, inertial effects ⁸¹, thermoelastic damping ⁸², and lattice friction ⁸³, drag from phonon viscosity and phonon scattering is the dominant drag mechanism on mobile dislocations under these material and strain rate conditions. When evaluated with inputs for copper at our strain rate of 10^7 s^{-1} this standard phonon drag strengthening model gives the green line in Fig. 16.

Although the first two contributions exhibit classical thermal softening, the third term has a large influence under high-rate conditions, which explains why the dynamic strength and hardness increase at our experimental strain rates. In fact, the increase predicted by the summation of the three terms are reasonably aligned in both absolute magnitude and in the scale of the change with temperature as shown by the black line in Fig. 16.

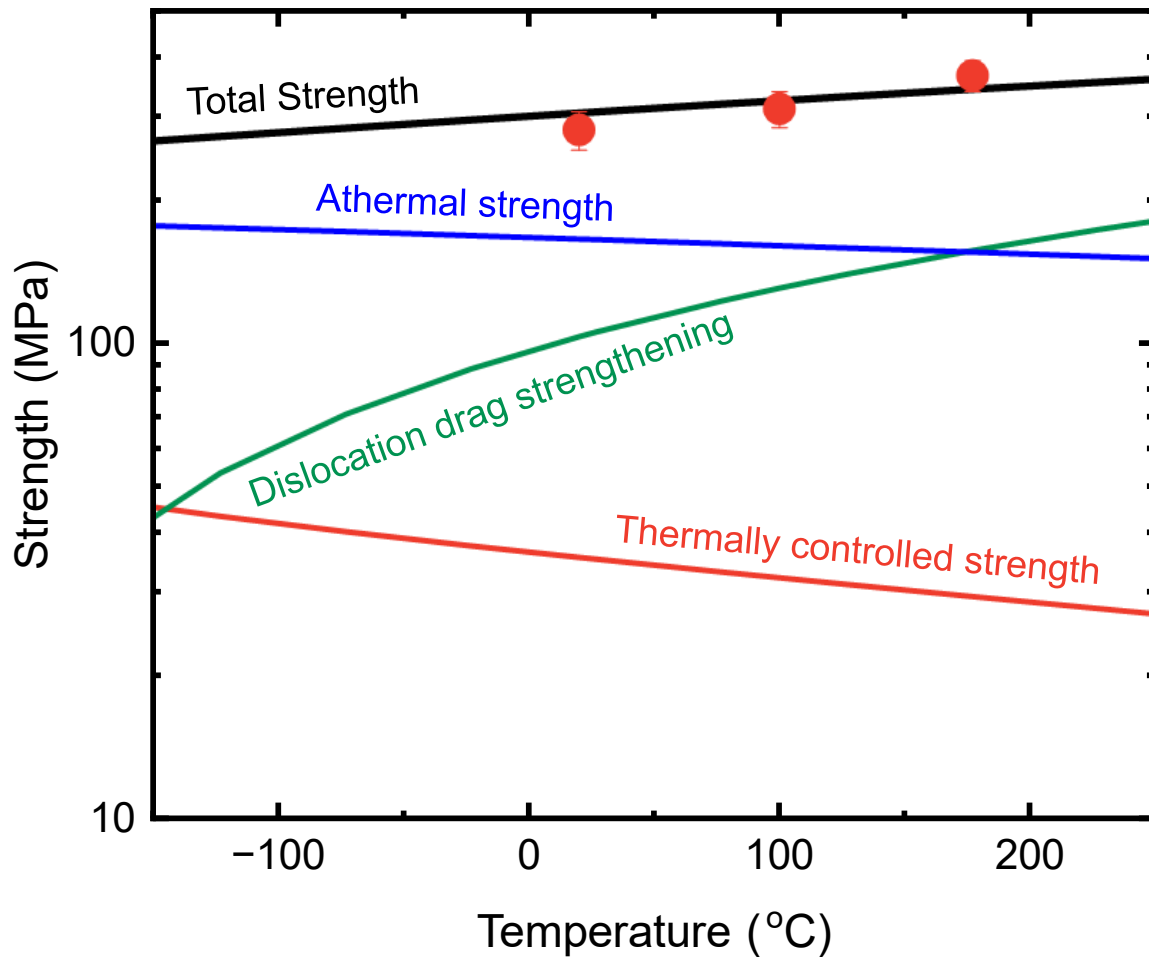


Figure 16: The contribution of each strength term for pure copper, calculated for a range of temperatures at a fixed strain rate of 10^7 s^{-1} . The magnitude of the increase in dislocation drag strengthening is greater than the magnitude of the decrease in all other strength terms, leading to a net increase in the total strength with temperature at this extreme strain rate. The experimental data points, plotted in red, are in good agreement with the total strength from all a summation of each strength component. Error bars on the experimental data points are the standard deviation of the strength measurements.

The unusual hotter-is-stronger trend in the present data are strictly an extreme-rate effect, and in fact the experimental data in Fig. 13b allude to a crossover point where the

temperature dependence of strength inflects; at low enough rates the data suggest a transition to classical thermally-activated behavior. This is elaborated more clearly in Fig. 17, where we reproduce data from Fig. 13b, and now include literature data for copper of similar purity ²⁰, tested at the closest lower strain rates where data is available ($\sim 2 \times 10^4 \text{ s}^{-1}$). The data at those lower rates exhibit the opposite temperature dependence from the present data. A clear transition regime exists between strain rates where those literature data end and our data begin, as seen in the crossover of the blue and red dotted schematic trendlines. The highlighted yellow band shows the range of strain rates where the crossover must occur. The strength equations described above can be used to evaluate the putative position of this crossover as well, by assessing the apparent activation energy for plasticity, Q_{app} :

$$Q_{app} = R \left(\frac{\partial \ln \dot{\epsilon}}{\partial \ln \sigma} \right)_T \left(\frac{\partial \ln \sigma}{\partial (1/T)} \right)_{\dot{\epsilon}} \quad (18)$$

where R is the gas constant. A theoretical value for the apparent activation energy for deformation can be determined from the strength equations above through the use of Eq. 18, and signals the tendency of the temperature dependence: positive values of Q_{app} suggest thermal softening due to thermal hopping of dislocations, while negative values are required to capture thermal hardening, as in our experiments, by virtue of a transition into a ballistic-like transport regime of dislocation motion ^{21,84}.

Arrhenius type plots for fixed strain rates can be calculated based on Eq. 13, 15, and 16 with the local slopes used to calculate Q_{app} . The theoretical prediction for Q_{app} is plotted for copper at a constant temperature of 20 °C on the secondary y-axis in Fig. 17. At a critical strain rate, Q_{app} changes from positive to negative, indicating a change in plasticity from thermally activated to ballistic transport of dislocations. However, for determining when phonon drag effects become significant, Q_{app} being negative is necessary but not alone sufficient for satisfying this condition. The derivative of the strain rate sensitivity parameter with stress at constant temperature, Eq. 19, must also be negative:

$$\left. \frac{\partial \left(\frac{\partial \ln \dot{\epsilon}}{\partial \ln \sigma} \right)}{\partial \sigma} \right|_T < 0 \quad (19)$$

In the present case, when Q_{app} crosses zero, the sensitivity parameter with stress is also negative, satisfying the phenomenological conditions for phonon drag to have a significant effect on strength. At any deformation rate beyond this threshold, thermal hardening dominates thermal softening due to phonon drag effects on moving dislocations.

Indeed, the model suggests a critical strain rate near 10^5 s^{-1} , where Q_{app} changes sign from positive to negative, signaling the change in controlling mechanism from thermally-activated softening to dislocation drag strengthening in the ballistic regime. This result provides interesting

new insight on prior work such as that in Refs. ^{85,86}, where thermal softening was observed in copper even at rates as high as $\sim 7 \times 10^4 \text{ s}^{-1}$; the present results clarify that only at even higher rates above 10^5 s^{-1} is the crossover expected.

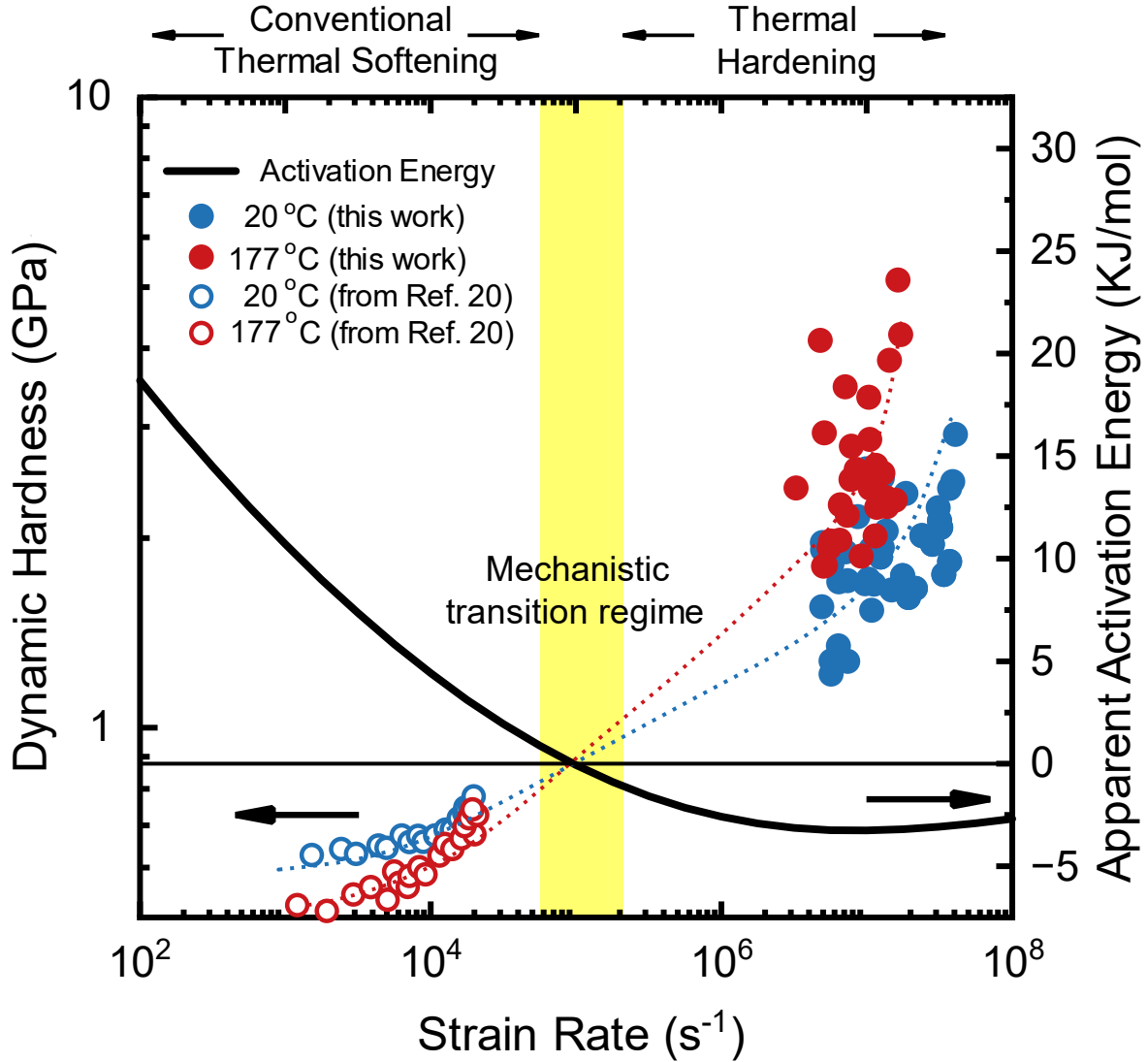


Figure 17: The blue and red dotted lines serve as a guide for the eye to connect hardness values, converted from strength values from Ref.²⁰ (empty circles), at low rates and dynamic hardness values from this work (filled circles). The apparent activation energy, Q_{app} , for plasticity (solid black line) of copper is also plotted as a function of strain rate at a constant temperature of 20 °C based on the model in the supplemental section. At rates below $\sim 10^5 \text{ s}^{-1}$, Q_{app} changes from positive, indicating thermally activated dislocation motion, to negative, one of the necessary conditions for phonon drag to be significant.

2.4: Implications and Conclusions

While models like those used here have been in the literature for many years, microballistic testing provides a new opportunity to quantitatively measure strength and hardness of materials at extreme strain rates, without complicating factors caused by strong shock physics. This not only provides a chance to test those models, as done here, but potentially to rapidly calibrate them for new materials and situations. The present data for three pure metals provides a starting point for focused study on those difficult-to-access mechanisms. The result that pure metals show significant thermal hardening at high strain rates also alludes to new strategies for materials design for extreme conditions; extrapolating conventional strength measurements to extreme conditions may lead not only to incorrect strength expectations, but even incorrect directional dependencies of those strengths. For example, while pure copper is a soft metal at low strain rates and would normally be expected to soften at high temperatures, the strength of copper measured here at 10^7 s^{-1} and 177°C is over 300 MPa; this is comparable to the conventional strength of 304 steel at this temperature ⁷⁶. The present results thus bolster the call for new modes of thinking about materials optimization for extreme conditions ^{87,88}.

Chapter 3: At extreme rates pure metals thermally harden while alloys thermally soften

The work in Chapter 2³⁹ experimentally showed that at these extreme strain rate deformations, above 10^6 s^{-1} , and low impact pressures, pure Cu, Au, and Ti increase strength with increasing temperature; they do not thermally soften, but rather thermally harden. This effect is due to kinetic limitations on defect motion and a change in deformation mechanism at extreme strain rates. The thermally activated transport of dislocations at lower rates gives way to ballistic transport of dislocations limited by dislocation-phonon interactions²¹. This transition leads to “anti-thermal” behavior where there is an increase in strength with increasing temperature, because temperature inflates the number of pinning points created by phonons. While this mechanism is dominant in high purity metals, it is always in competition with other deformation mechanisms, and it is not yet clear what its range of dominance will be in the face of, e.g., added alloying elements, which introduce new strengthening mechanisms not present in pure metals. At low strain rates, alloying leads to additional dislocation pinning mechanisms; at the highest strain rates where phonon drag is relevant, however, such alloying effects are not well quantified. Here, using microparticle impact testing, we systematically explore several orders of magnitude of solid-solution additions into pure metals. We show that phonon drag crosses from dominant to secondary as a rate-limiting mechanism as the solute level rises: Only high purity metals strengthen with increasing temperature, and there is a clear crossover to thermally activated flow at lower purity.

Nickel rods with nominal purity levels of 99.999%, 99.995%, 99.95%, 99.5% and 99% (ThermoFisher, USA) were annealed at 500 °C for 7 hours before being ground flat and polished

to a mirror finish using standard metallographic techniques. Alumina microparticles (Cospheric, Santa Barbara, CA) $20 \pm 1 \mu\text{m}$ were used as indenters for impact experiments due to their high hardness compared to the nickel substrates (~ 15.7 vs. ~ 1.15 GPa at quasistatic rates)^{64,89}. 99.99% Ti (Alfa Aesar, USA) and a Grade 2 CP-Ti plate (Online Metals, USA) were annealed at 600 °C for 10 hours then ground and polished to a mirror finish. 99.999% and 99.95% Au rods (ThermoFisher) were rolled to an 85% reduction in thickness before being annealed at 125 °C for 3 hours. Note that the high purity samples of Ti and Au are the same materials used in our prior work³⁹, and we reproduce the data for those samples here, alongside new data for the alloyed samples of Ti and Au.

3.1: Extreme Strain Rate Testing on Ni Alloys

Impact experiments were carried out using high velocity microparticle impacts with the laser-induced particle impact test (LIPIT), detailed in Chapters 1 and 2. The test materials were nickel substrates of five different purities, 99.999%, 99.995%, 99.95%, 99.5% and 99%, with compositions given in Table 1; the solutes in these Ni samples are predominantly C, Fe, Cu, and Mn. These samples were tested at extreme strain rates by impacting them with hard alumina microspheres, $20 \pm 1 \mu\text{m}$ in diameter, at two different temperatures, 20 and 155 °C. The velocities of our experiments ($v_i = 60\text{-}270$ m/s) were chosen to be well below the regime where shock effects (such as jetting, hydrodynamic penetration, etc.) occur (~ 500 m/s⁶²). For each material, the impact velocity, v_i , is plotted against the coefficient of restitution, CoR, or ratio of rebound to impact velocity (v_r/v_i), on a double logarithmic scale, Fig 18. In these experiments, the alumina impactors are effectively undeforming (elastic); higher CoR signifies that more energy is returned

to the particle after impact and less is consumed as plasticity in the substrate. At the two different temperatures, for impacts at the same velocity, a faster rebound is a first indication of a higher strength metal.

Table 1: The composition of each alloy given in atomic percent. The 99% Ni alloy met or exceeded the physical and chemical requirements for ASTM B160. The substitutional elements (Mn, Cu, Fe, Si, and Co) have very little impact on alloy strength, while the interstitial elements (C, B, and S) have strong interactions with dislocations; their total composition in the final column is used in the model.

Name	Ni	C	Mn	Cu	Fe	Si	S	B	Co	Total Interstitial
99.999	99.997	0.003	1.07×10^{-7}	1.29×10^{-5}	2.5×10^{-4}	6.3×10^{-6}	-	-	-	0.003
99.995	99.986	0.01	2.7×10^{-6}	7.4×10^{-6}	0.001	8.3×10^{-6}	0.0004	-	-	0.104
99.95	99.95	0.037	-	-	-	-	0.003	0.005	0.001	0.045
99.5	99.36	0.244	0.19	0.018	0.01	0.17	3.7×10^{-4}			0.24437
99*	99	Up to 1.0	0.35 (Max)	0.25 (Max)	0.40 (Max)	0.35 (Max)	-	-	-	

For a fixed temperature as the impact velocity is increased, more energy is dissipated as plasticity, as seen through a drop in CoR with increasing impact velocity. The parabolic scaling of the data in Fig. 18 is an expected form for plastic impacts, following the model of Wu et al. ^{60,61} for a contact with a single yield stress:

$$CoR = \frac{v_r}{v_i} = 0.78 \left(\frac{\sqrt{26} \cdot \sigma_y^{3/2}}{E^* v_i \sqrt{\rho}} \right)^{1/2} \quad (20)$$

where σ_y is the dynamic yield strength, E^* is the effective modulus between the impactor and substrate, and ρ is the density of the impactor. The data in Fig. 18 can be fitted with Eq. 20 with a single parameter, σ_y . This value provides a measure of the average strength of the substrate over

the duration of the impact. The fits are shown as solid lines in Fig. 18 for each set of data, along with the fitted dynamic yield strengths.

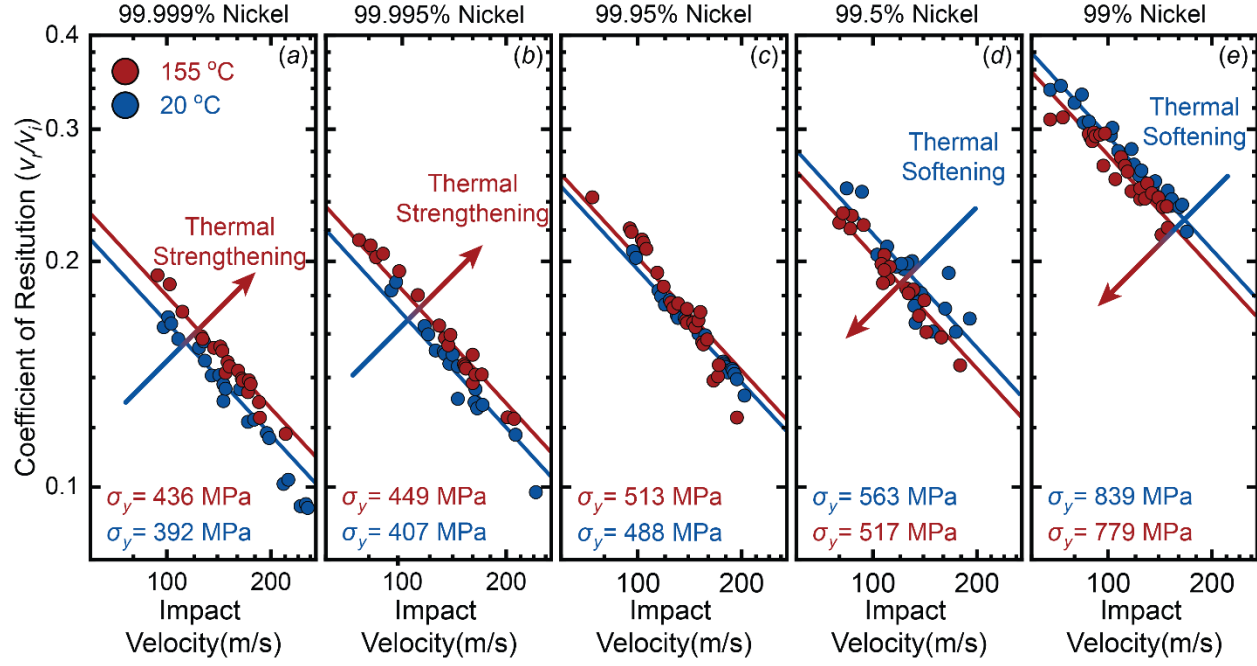


Figure 18: The impact velocity versus coefficient of restitution is plotted on a double logarithmic scale for **(a)** 99.999% Ni **(b)** 99.995% Ni **(c)** 99.95% Ni **(d)** 99.5% Ni and **(e)** 99% Ni at 20 °C and 155 °C. There is a clear change from thermal hardening to thermal softening when increasing the solute content in Ni with 99.95% Ni serving as the approximate transition point. Solid lines indicate the parabolic scaling of the impact and rebound behavior as proposed by Wu et al. This scaling was used to calculate the dynamic yield strength σ_y for each material at each temperature.

There are two interesting trends in Fig. 18. First, alloying increases strength as expected; comparing across the datasets, it is clear that as the purity level of the Ni decreases from left to right across the panels in the figure, the response curves rise and the corresponding fitted value of σ_y increases. However, closer inspection reveals that this alloy strengthening effect is more potent at lower temperatures than at higher ones, meaning that the blue curves rise faster than

the red ones in Fig. 18. In Fig. 18a-c a clear thermal strengthening, “hotter-is-stronger”, trend is seen, where impacts rebound faster and the calculated σ_y values are larger at the hotter test temperature. However, at the higher alloying levels (i.e., lower purity levels), the CoR curves overlap more clearly, and the strength values at the two temperatures become close. In Fig. 18d-e, a new trend is seen where the test carried out at higher temperatures now leads to thermal softening, “hotter-is-softer”, again shown both as lower rebound velocities and lower calculated σ_y values.

For each particle impact event, in addition to the in-situ observations of the particles in flight, a plastic indentation is left in the substrate. Each indentation is observed and measured using 3D laser scanning confocal microscopy. Across all test materials and temperatures, larger indentation volumes are a first indication of a softer material (i.e., more deformation). In Fig. 19, across the range of alloys, we see complementary trends in indentation volume to those observed earlier with σ_y . First, the addition of alloying elements leads to smaller indentations (hardening) across the series horizontally. Second, however, the higher purity samples have smaller indentation volumes with increasing temperature, and the lower purity samples have larger indentation volumes with increasing temperature. The two indentation volumes in Fig. 19e at the two test temperatures are approximately the same, consistent with the overlap seen in the corresponding CoR curves; this is roughly the point of the crossover from “hotter-is-stronger” to “hotter-is-softer” with the addition of alloying elements.

The impact indentation can be further used as the quantitative basis of the dynamic hardness, H_d , assessed as:

$$H_d = \frac{1/2 \cdot m_p \cdot (v_i^2 - v_r^2)}{V} \quad (21)$$

where V is the indentation volume, and m_p is the mass of the impactor, measured using the density of the particle and its measured diameter. The dynamic hardness for each impact is plotted as a cumulative probability in Fig. 20, with the average strain rate across all impacts for each material and temperature in Table 2.

Table 2: The average strain rate and standard deviation for impacts onto each material.			
	Temperature	Average Strain Rate ($\times 10^6$)	Standard Deviation ($\times 10^6$)
99.999% Ni	20 °C	7.77	2.14
	155 °C	7.61	1.56
99.995% Ni	20 °C	7.29	1.85
	155 °C	6.73	1.83
99.95% Ni	20 °C	7.69	1.61
	155 °C	6.86	1.36
99.5% Ni	20 °C	6.57	1.41
	155 °C	5.92	1.30
99% Ni	20 °C	5.96	1.47
	155 °C	5.68	1.21

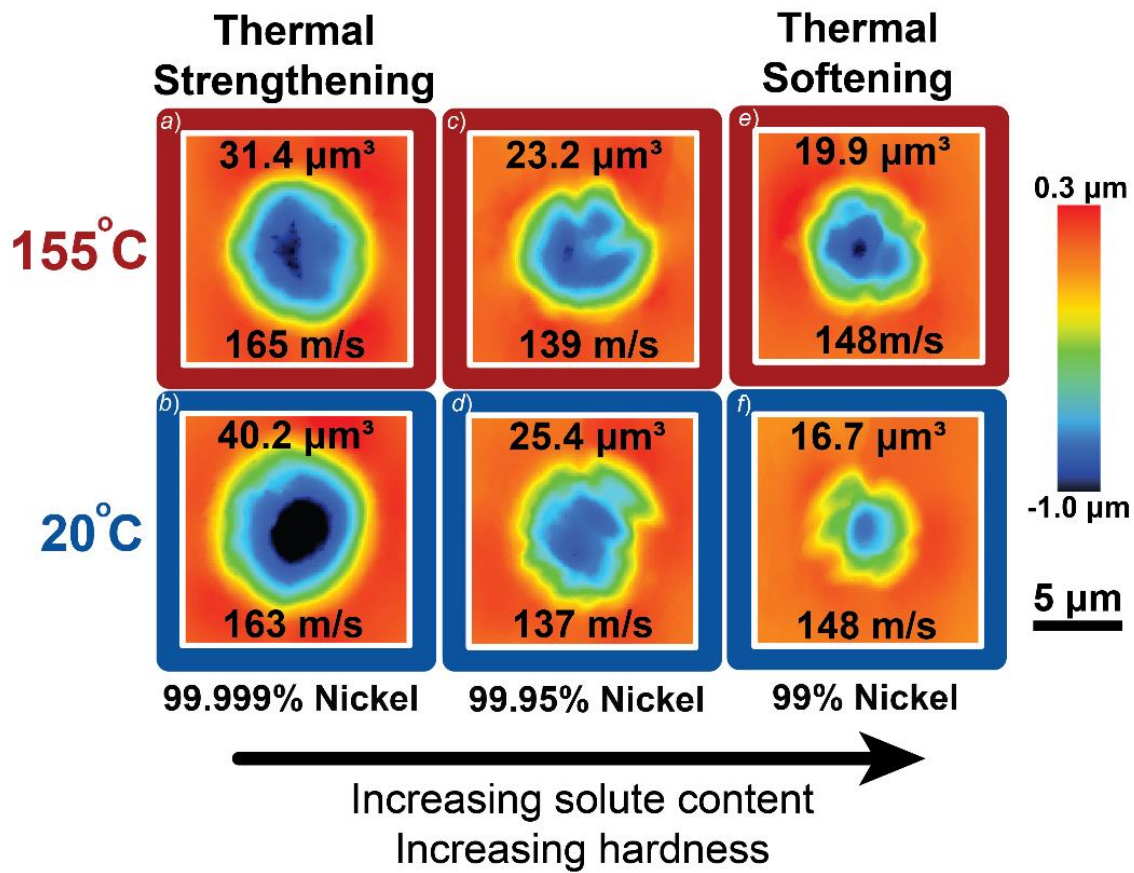


Figure 19: Laser scanning confocal micrographs are shown for (a,b) 99.999% Ni, (c,d) 99.95% Ni, and (e,f) 99% Ni with inserts showing the measured crater volume and impact velocities. The crater volume increases from (a) to (b) indicating the expected thermal hardening in pure metals. The crater volume decreases from (e) to (f) indicating classic thermal softening in the lowest purity material. The crater volumes in (c) to (d) are approximately equal corresponding to the transition point from thermal hardening to thermal softening. What is more, as the purity level decreases, the crater volume decreases, another expected form of plasticity where alloy elements increase the hardness of a metal.

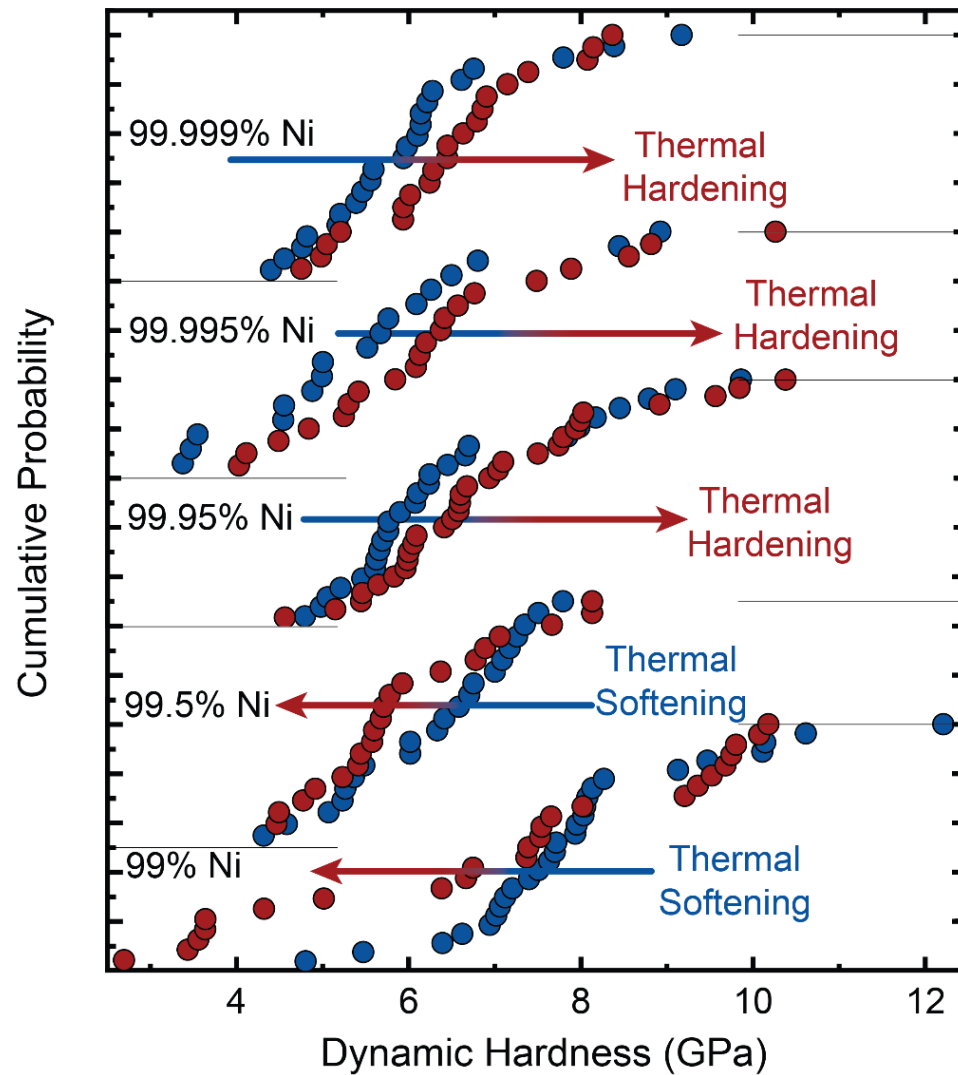


Figure 20: The dynamic hardness for each impact experiment was calculated and plotted as a cumulative probability for each test material and temperature. As with the dynamic yield strength and measured crater volumes, the highest purity samples exhibited thermal hardening and the lowest purity samples exhibited thermal softening. Additionally, The average hardness at 20 °C is increasing with increasing solute content, the expected form of plasticity.

As with the CoR and σ_y in Fig. 18, the hardness values in Fig. 20 show the same two major trends. First, alloying leads to higher hardness generally, as the cumulative distributions shift to the right with decreasing purity. Second, however, we see the same thermal crossover: Fig. 20 a-c show that the higher purity Ni is thermal hardening (hotter samples harder than cooler ones,

red curve to the right of blue), while the lower purity Ni, Fig. 20d,e, is thermally softening, indicated by a flip in the trend of hardness data (blue curve to the right of red).

3.2: Deformation mechanisms including solute effects

The strength of a metal at strain rates above 10^6 s^{-1} can be approximated as the linear combination of the available strengthening mechanisms. As note in Chapter 2, in pure metals these include the thermally activated motion of dislocations at the intrinsic lattice strength (Peierls barriers, σ_{th}) and athermal strengthening from dislocation-grain boundary and dislocation-dislocation interactions (σ_a)^{21,39,43,73,90}; both of these classical mechanisms exhibit thermal softening.

The thermal strengthening mechanism comes from dislocation interactions with thermally activated barriers in the crystal lattice: i.e. internal lattice frictions, Peierls barriers, and are fully overcomable by thermal fluctuations alone – leading to a decrease in strength with increasing temperature:

$$\sigma_{th} = \left[1 - \left(\frac{kT}{g_0 \mu_T b^3} \ln \left(\frac{\dot{\epsilon}_0}{\dot{\epsilon}} \right) \right)^{1/q} \right]^{1/p} \sigma_0 \quad (22)$$

$$\mu_T = \mu_o - \frac{s}{e^{\left(\frac{T_o}{T}\right)} - 1} \quad (23)$$

In Eq 22, σ_o is the stress needed to overcome the short-range barriers at 0 K and $q = 2/3$ and $p = 2$ are fitting parameters based on the energy shape of the barriers. The temperature dependent shear modulus, Eq. 23, can be calculated using the model by Varshni⁹¹, where T_o and s are constants equal to 258 K and 8.839 GPa and μ_o is the shear modulus at 0 K, 84.52 GPa, for pure Ni.

Dislocations also interact with other dislocations and grain boundaries, providing athermal strengthening contributions. The magnitude increase in strength of these interactions is much greater than with short-range barriers and, importantly, cannot be overcome by thermal fluctuations alone.

$$\sigma_a = \frac{\alpha_G \mu_T \sqrt{b}}{\sqrt{D}} + \alpha_{disl} \mu_T b \sqrt{\rho} \quad (24)$$

The interaction of dislocations with grain boundaries is modeled here using a simple Hall-Petch like relation, the first term in Eq. 24, where D is the grain size after impact, estimated to be on the order of several hundred nanometers from the work of Ref.⁷⁴, and α_G is an interaction parameter between dislocations and grain boundaries taken to be 0.15. The second term of Eq. 24 captures the strengthening from dislocation-forest and dislocation-dislocation interactions through the mobile dislocation density, ρ ($\sim 10^{13} \text{ m}^{-2}$), and the dislocation-dislocation interaction parameter $\alpha_{disl} = 0.5$. While these barriers cannot be overcome by thermal fluctuations alone,

there is a small decrease in strength in each term with temperature due to a decrease in the temperature dependent shear modulus.

Additionally, there is a dislocation drag term that arises from dislocations interacting with phonons:

$$\sigma_d = \frac{B_d \dot{\epsilon}}{\rho_m b^2} \quad (25)$$

where b is the burgers vector, B_d is the drag coefficient, $\dot{\epsilon}$ is the strain rate, and ρ_m is the mobile dislocation density. Notably, the drag term of Eq. 25 has antithermal behavior, i.e., stronger-is-hotter behavior, because B_d increases linearly with temperature^{46,73,77,78}. At the very high strain rates of our tests $\sim 7 \times 10^6 \text{ s}^{-1}$, the dislocation drag strengthening term of Eq. 25 can be dominant over the others in pure metals, giving rise to an overall increase in strength with increasing temperature³⁹. This was quantitatively established in our prior work on pure Cu, Au, and Ti with similar LIPIT experiments in³⁹, and the results in Figs. 18a and 20 extend the observation of thermal hardening to pure Ni as well; thermal strengthening is prima facie evidence that dislocation drag is the dominant mechanism in our experiments on pure Ni.

However, the addition of solute elements into a Ni matrix introduces a new strengthening mechanism that can both increase the overall magnitude of the strength and have a different scaling dependency with temperature. Solute atoms provide new “soft” pinning points for dislocations to interact with, which, like Peierls barriers, can be overcome by suitable thermal

fluctuations. Solute strengthening is represented by a fourth additive contribution (σ_s) to the total strength σ_y :

$$\sigma_y = \sigma_{th} + \sigma_a + \sigma_d + \sigma_s \quad (26)$$

The role of solute strengthening has been addressed by many authors ⁹²⁻⁹⁴, and the present alloys are relatively dilute, and primarily incorporate interstitial elements (Table 1) such as C. While there are other solute elements in Ni, they contribute to substitutional strengthening at the negligible level of a few MPa, by comparison to interstitial elements which have a much larger effect due to their anisotropic lattice distortions. For such materials, the expression developed by Follansbee et al. ⁹⁰ is appropriate for an illustrative analysis, and was calibrated by those authors to the specific case of Ni containing interstitial C:

$$\sigma_s = \left(\left[1 - \left(\frac{kT}{g_0 b^3 \mu_T} \ln \left(\frac{\dot{\epsilon}_0}{\dot{\epsilon}} \right) \right) \right]^{3/2} \sigma_c c \right)^n \quad (27)$$

Here g_0 is the normalized total free energy, k is the Boltzmann constant, b is the burgers vector, $\dot{\epsilon}_0 = 10^{10} \text{ s}^{-1}$ is a reference strain rate, and μ_T is the temperature-dependent shear modulus. The exponent $n = 1.4$ speaks to the spacing of solutes in the path of dislocations. The threshold stress, $\sigma_c = 20 \text{ GPa}$ according to Follansbee et al. ⁹⁰, and c is the interstitial solute content.

Because this newly available strengthening term is positive and additive to the others, and yet is thermally controlled, it has two major effects: it strengthens, but it lowers the temperature dependence of strength (note that $d\sigma/dT$ is negative in Eq. 27). As a result, pure samples, which are dominated at these high strain rates by dislocation drag (σ_d), have the highest anti-thermal hardening and the new σ_s term is not critical to that temperature dependence. As solutes are added to the system, though, σ_s increases its contribution, and the anti-thermal hardening slope is lessened. In the lowest purity samples, the slope has indeed flipped sign, reflecting a crossover to dominance of the σ_s term, including its prevailing thermal activation characteristics.

The mechanistic cross-over described above can be seen in fuller form by evaluating Eq. 26, with the various strengthening terms populated with data for Ni as laid out above and in the supplemental materials. Figure 21 plots the experimentally assessed σ_y values at the experimental strain rate of 10^7 s^{-1} as well as Eq. 26 using the full interstitial solute content for each alloy from Table 1. Both of the critical experimental trends are well captured by the model. First, solute elements cause strengthening at all rates, and the rise follows the expected scaling of Eq. 26 reasonably well. Second, the temperature dependence of strength changes as the solute drag term becomes dominant at higher solute concentrations. There is a critical concentration of solutes where, at extreme strain rates, the dominant mechanisms cross-over, and the strength of Ni becomes nominally independent of temperature. This crossover point lies at $c \sim 0.4\%$ for interstitials in Ni. This concentration also marks the point where Ni alloys transition from antithermal “hotter-is-stronger” to more typical thermally-activated “hotter-is-softer” deformation.

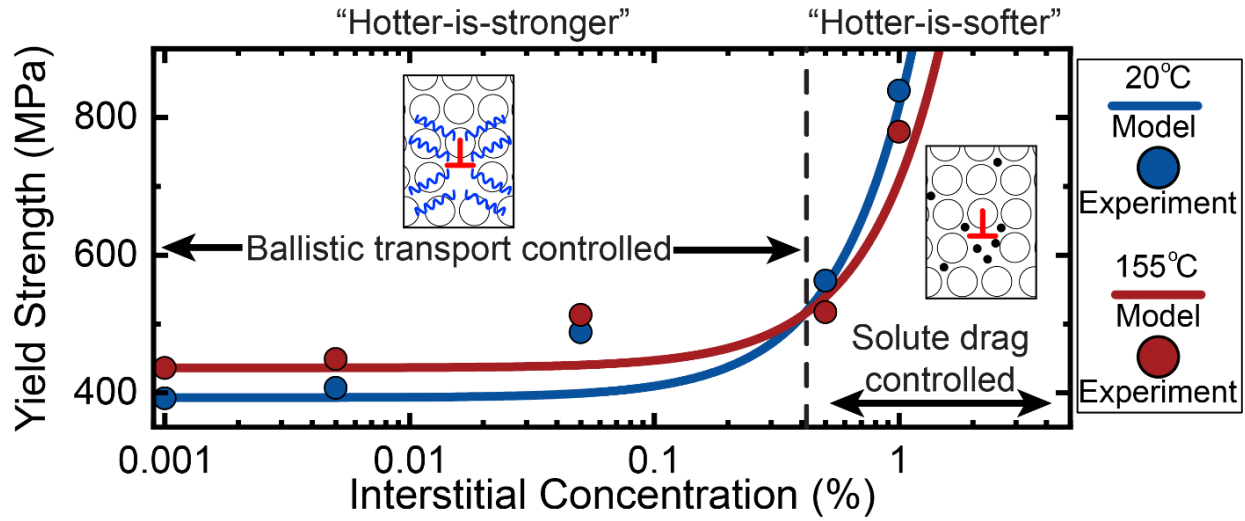


Figure 21: The experimentally assessed dynamic yield strength for each test material is plotted as a function of interstitial concentration, filled circles. As the interstitial concentration increases, for each temperature, the yield strength increases. Solid lines show the total yield strength predicted from the model, including the solute strengthening term from Eq. 27. As the interstitial concentration increases from 0.001%, the materials exhibit thermal hardening then cross over to thermal softening at an interstitial concentration of $\sim 0.4\%$. This cross over from “hotter-is-stronger” to “hotter-is-softer” arises from a change in the dislocation kinetics from ballistic transport of dislocations to solute drag controlled, which is thermally activated.

The transition we see in Fig. 21 is one that is expected to be common to many materials, because the competition amongst deformation mechanisms in Eq. 26 is common physics, although the details of the specific mechanisms and their functional forms may vary. For metals, the trend in Fig. 21 is broadly expected to prevail, and in Figs. 22 and 23 we demonstrate the generality of the present observations for two additional alloy families based on Ti and Au.

Two test materials of Ti were chosen, high purity 99.99% Ti and commercially pure (CP-Ti), Table 3. Data for 99.99% Ti was taken from our recent work in Ref. ³⁹ and adapted here.

Coefficient of restitution plots for alumina impactors on each test material are shown in Fig. 22 a,b. Fitting the impact and rebound data to the Wu et al. plasticity model ⁶⁰, as expected, at 20 °C, 99.99% Ti has a lower dynamic strength than CP-Ti, 776 MPa compared to 1029 MPa. However, shown in our previous work ³⁹, the 99.99% Ti sample thermally strengthens by 15% when heated to 200 °C while the CP-Ti sample thermally softens by 10% under the same impact conditions. This same trend is seen in Fig. 22c with the dynamic hardness cumulative probabilities. As with the strength, the hardness for 99.99% Ti is lower than the hardness for CP-Ti and exhibits thermal hardening, while the hardness for CP-Ti shows a clear thermal softening trend.

Table 3: The atomic percentages of the high purity Ti rod and Grade 2 commercially pure Ti plate are given below.						
	Ti	N	C	H	Fe	O
High Purity Ti	99.82	6.8×10 ⁻⁴	0.029	0.016	7.6×10 ⁻⁴	0.132
CP-Ti Grade 2	99.3	0.03	0.1	0.015	0.3	0.25

As with the Ti, Au with two different compositions were chosen as test materials, 99.999% Au and 99.95% Au, Table 4. Data for the 99.999% Au material was again taken from our recent work Ref.³⁹, and adapted for this work. In Figure 23 a,b, the higher purity Au sample exhibited thermal strengthening while the lower purity sample thermally softened when using the Wu et al. plasticity model for dynamic strength ⁶⁰. This trend is again reaffirmed using the dynamic hardness measurements and crater volumes. Because the temperature different is smaller for these samples, only ~80 °C, there is less of a shift in the hardness and strength when compared

to the other materials. Additionally, when the Au samples were prepared, the gold rods were cold rolled prior to annealing. This cold rolling added more defects into the material which increases the overall strength.

What is more, despite rolling to the same reduction in thickness, the work hardening rate of the two samples is expected to be different because of the solute elements present in the 99.95% Au sample. From Lucas et al. it has been shown that work hardening has a significant effect on the rebound behavior and crater volume size under high velocity microparticle impacts.

Table 4: The composition of each Au test material is given below in atomic percent.						
	Au	Ag	Ca	Cu	Ti	Pt
99.99% Au	99.999	7.1×10^{-4}	3.9×10^{-4}	6.2×10^{-5}	4.5×10^{-4}	4.0×10^{-5}
99.95% Au	99.97	9.8×10^{-3}	2.5×10^{-3}	0.015	-	0.001

In both cases, only the highest purity materials thermally hardened while the alloyed materials thermally softened. Because both of these materials exhibited the same behavior, one can infer that the “hotter-is-stronger” to “hotter-is-softer” transition seen is not due to the crystal structure (HCP Ti versus FCC Ni) or any native oxide growth on the surface, as Au is noble does not form a surface oxide under these conditions.

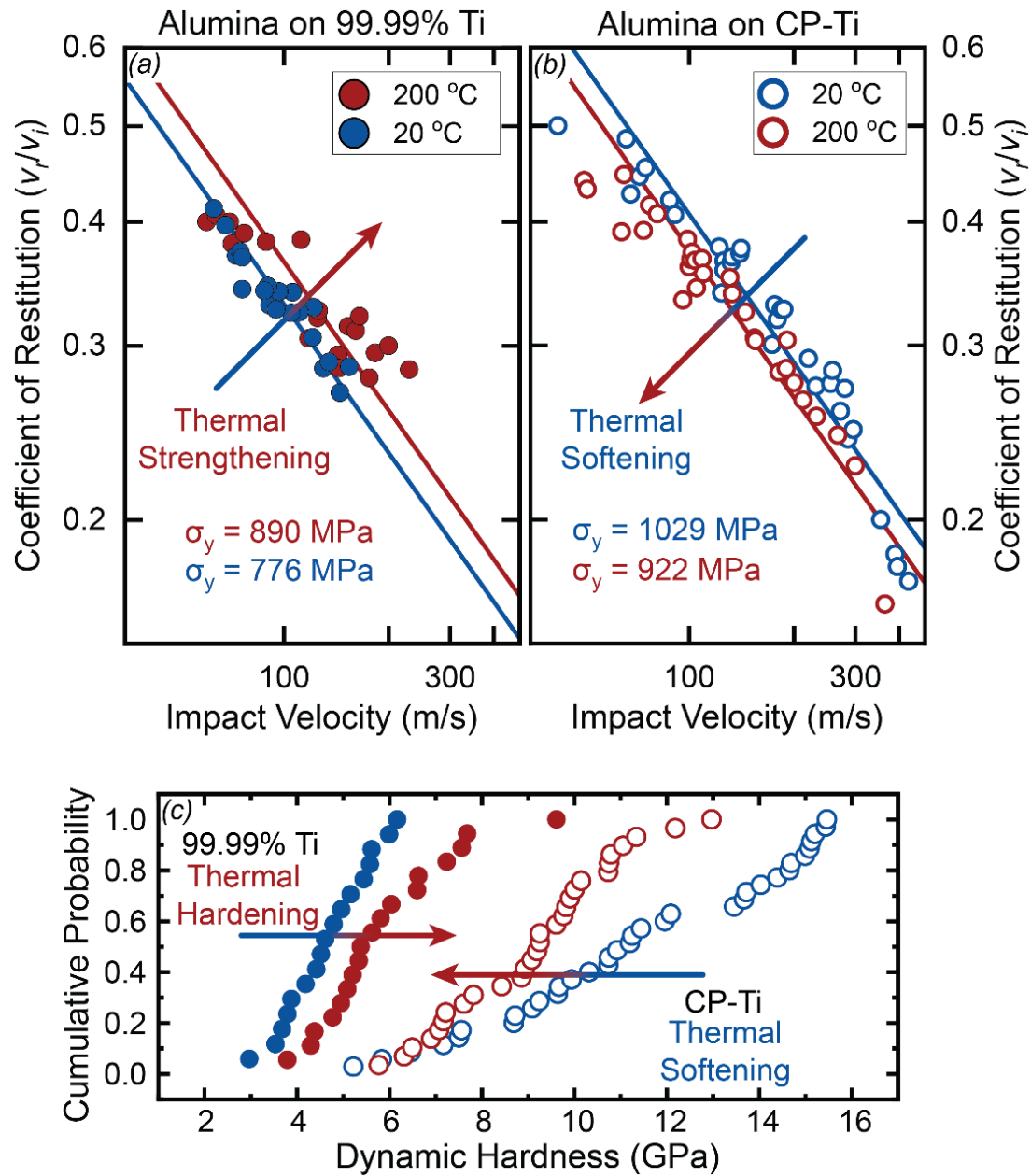


Figure 22: The coefficient of restitution is plotted as a function of impact velocity for **(a)** 99.99% Ti and for **(b)** CP-Ti, at 20 °C and 200 °C. 99.99% Ti thermally strengthens while CP-Ti thermally softens due to the increased number of pinning points for dislocations from the higher solute content. Solid line insets over the data show the parabolic scaling of the impact from the plasticity model by Wu et al. The dynamic yield strengths are calculated for each sample. The dynamic hardness of each test material is plotted as a cumulative probability in **(c)**. As with the dynamic yield strength, the dynamic hardness increasing with increasing temperature for the 99.99% Ti samples and decreases with increasing temperature for the CP-Ti samples. Data for 99.99% Ti was taken from our previous work in Ref. ³⁹.

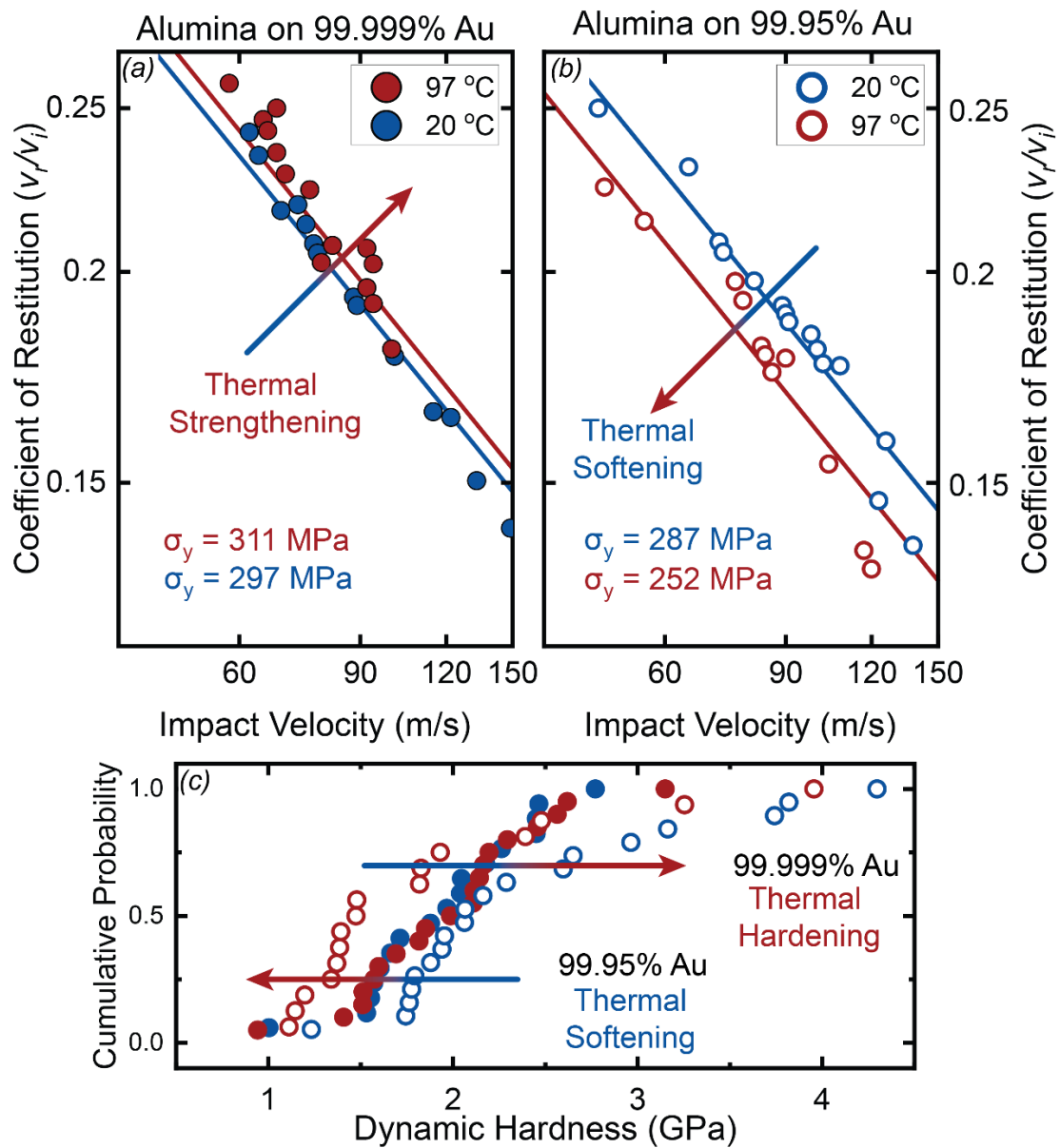


Figure 23: The coefficient of restitution is plotted as a function of impact velocity for **(a)** 99.999% Au and for **(b)** 99.955 Au, at 20 °C and 97 °C. 99.999% Au thermally strengthens while 99.95% thermally softens. Solid line insets over the data show the parabolic scaling of the impact from the plasticity model by Wu et al. The dynamic yield strengths are calculated for each sample. The dynamic hardness of each test material is plotted as a cumulative probability in **(c)**. While the hardness data for 99.999% Au shows significant overlap, consistent with the coefficient of restitution plot, the trends still hold where the high purity sample is thermally hardening while the lower purity sample is thermally softening. A lower maximum test temperature was used because the Au samples were work hardened prior to impact experiments so recrystallization of the work hardened samples was avoided.

3.3: Summary and Implications

In summary, although models for high-rate ($\sim 10^4 \text{ s}^{-1}$) deformation mechanisms in metals have existed for several decades, LIPIT accesses extreme rates, beyond 10^6 s^{-1} , without strong shock effects dominating the material response. In addition to clean observations of phonon-drag dominated deformation, with their “hotter-is-stronger” behavior, here we illustrate the ability of LIPIT to map the range of dominance of that mechanism as it crosses over to other more conventional thermally-activated deformation mechanisms. Above strain rates of 10^5 s^{-1} , highly pure metals increase strength and hardness with increasing temperature, while lower purity Ni thermally softens. This transition point from drag-controlled to solute-pinning-controlled mechanisms conforms to mechanistic strength models that account for both the strain rate and temperature effects of dislocation interactions with phonons as well as dislocation interactions with interstitials. There are many technological applications where such mechanistic transitions are of clear relevance, including hard particle erosion and sandblasting, high-speed subtractive manufacturing, cold dynamic gas spray manufacturing, and hypersonic transport. The present results suggest that when designing for, or protecting against, deformation at these rates, it is critical to consider all the mechanisms at play. Clearly, alloying can be used as a tool to precisely manipulate the temperature dependence of strength in unusual ways at these high rates, and there are compositions such as the $c \sim 0.4\%$ composition in Ni seen in Fig. 21 where the mechanical response is effectively temperature independent. We look forward to progress in the design of alloys for extreme conditions in light of the unusual deformation physics that prevail in microparticle impacts.

Chapter 4: Breakdown of the “hotter-is-stronger” trend close to the melting temperature

Microballistic impact experiments produce strain rates between 10^6 – 10^9 s⁻¹ at pressures typically well below about 10 GPa in metals, and below the onset of where hydrodynamic-effects dominate the material response in metals. As a result, these methods have successfully quantified increases in metals dynamic strength and dynamic hardness with increasing strain rates^{32,39,40}, in conditions that are cleanly and demonstrably measuring the mechanical response of the material.

The outcomes from our work in Chapters 2 and 3 using microparticle impact experiments³⁹ is a quantitative documentation of the thermal-hardening, ballistic transport regime in several metals, including Cu, Au, Ti, and Ni. In those metals, heating by up to 177° C led to significant (~30%) increases in high strain-rate strength and hardness, a result of the mechanism transition into the ballistic regime where dislocation motion is governed by phonon interactions rather than obstacle bypass. In addition to being counterintuitive, this “hotter-is-stronger” behavior begs an immediate question: in what manner does the ballistic transport mechanism break down at higher homologous temperatures? Clearly metals cannot be expected to harden at indefinitely higher temperatures as they approach their melting point, but the transition of deformation behavior from antithermal hardening to thermal softening has not, to our knowledge, been quantitatively established before. The purpose of the present work is to provide a first systematic exploration of that topic, by focusing on a low melting point metal—tin—and establishing both

thermal hardening and thermal softening regimes in a single system with a single test method, at extreme strain rates $\sim 10^7 \text{ s}^{-1}$.

Tin substrates (from Alfa Aesar, Ward Hill, MA) were cut from a plate using a water jet, and ground and polished using standard metallographic techniques to a mirror finish. 304 stainless-steel microparticles (Cospheric, Santa Barbara, CA), $12.5 \pm 1 \text{ }\mu\text{m}$, were used as indenters during impact experiments due to their high static hardness compared to tin (0.645 GPa vs. 0.03 GPa), converted from yield strength values.

4.1: Quantifying High-Rate Properties of Tin

Impact experiments were carried out using optically driven microballistics, detailed in Chapter 1, where hard 304 stainless steel particles were accelerated to velocities ranging from 30–1100 m/s onto tin substrates. The data in Fig. 24a show the particle rebound response of each impact test, plotted on a double logarithmic scale as the impact velocity, v_i , versus the coefficient of restitution (CoR), which is the ratio of the rebound velocity to impact velocity, v_r/v_i . Because the particles (hardness $\sim 123 \text{ kgf/mm}^2$) are far harder than the target ($\sim 3.9 \text{ kgf/mm}^2$), we assume that for most all impacts in this work the deformation partitions fully to the substrate. Thus, higher CoR values indicate that more energy is returned to the impacting particle, and less energy is consumed as plasticity in the substrate. Across the range of impact velocities and temperatures tested, CoR drops with velocity as seen in Fig. 24a, as more plastic dissipation occurs at higher speeds. The full set of data can best be understood by considering it in two

halves, at lower (< 500 m/s) and higher (>500 m/s) velocities. We take these regimes in turn in what follows.

The data at lower velocities is isolated in Fig. 24b, and follows an expected power-law form for plastic impacts based on the work of Wu et al.^{60,61}. In this range all of the impacts are past the elastic-plastic transition (v_y)^{30,62,95}, which occurs below 1 m/s and plasticity dominates the response:

$$CoR = \frac{v_r}{v_i} = \alpha \left(\frac{v_y}{v_i} \cdot \frac{E^*}{Y_d} \right)^{1/2} \quad (28)$$

Here Y_d is the dynamic yield strength, E^* is the effective modulus between the impactor and substrate, and $\alpha = 0.78$ is a prefactor for elastic-perfectly plastic impacts under an elastic impactor⁶². While a high velocity particle impact is very dynamic in nature, this simple scaling relation matches well with the rebound behavior of the particle. Moreover, the data can be fitted with a single parameter, the dynamic yield strength, Y_d , which provides an average strength measure over the duration of the impact.

Fitting the data in Fig. 24b with Eq.28 reveals that the effective dynamic yield strength of tin increases from 220 to 292 MPa with a temperature increase from 20 to 150 °C. While this antithermal “hotter is stronger” behavior is counterintuitive, it is indeed expected at these rates due to dislocation motion being controlled by ballistic transport; dislocation–phonon interactions increase the drag strength with temperature³⁹. As the impact velocity is increased further, however, the data in the high velocity regime no longer follow the parabolic scaling law between the impact velocity and CoR, as seen in Fig. 24c. Instead, we see a more rapid decline in CoR, a

divergence that has been seen previously at room temperature in tin and shown to be associated with the onset of melting^{30,32}. Such melting has been analyzed in those works; it is not the result of pressure, shock, or shock-release, but rather of extensive plasticity, which in turn generates adiabatic heat enough to melt the target. Because excess energy from the impact is consumed by this softening mechanism, less is returned to the particle in the form of rebound velocity, steepening the slope in Fig. 24a. Fig. 24c isolates this higher velocity regime, and what stands out when compared to Fig. 24b is that in this high velocity range, the CoR drops with increasing temperature: in the regime of plastic impacts the substrate gets stronger with temperature, but once the transition to melting has been reached, “hotter-is-softer”. This is the first direct observation of a transition from anti-thermal to thermally-activated deformation at high rates and temperatures.

The transition from “hotter-is-stronger” to “hotter-is-softer” in Fig. 24 is reaffirmed with a second strength measurement that is independent of the Y_d fitting, namely the dynamic hardness:

$$H_d = \frac{1/2 \cdot m_p \cdot (v_i^2 - v_r^2)}{V} \quad (29)$$

which is the energy dissipated as plasticity in the substrate normalized by the impact crater volume, V , with m_p the mass of the impactor calculated using the measured particle diameter and density. The impact and rebound velocities are measured from the in-situ image sequence and the crater volume is measured using 3D laser scanning confocal microscopy.

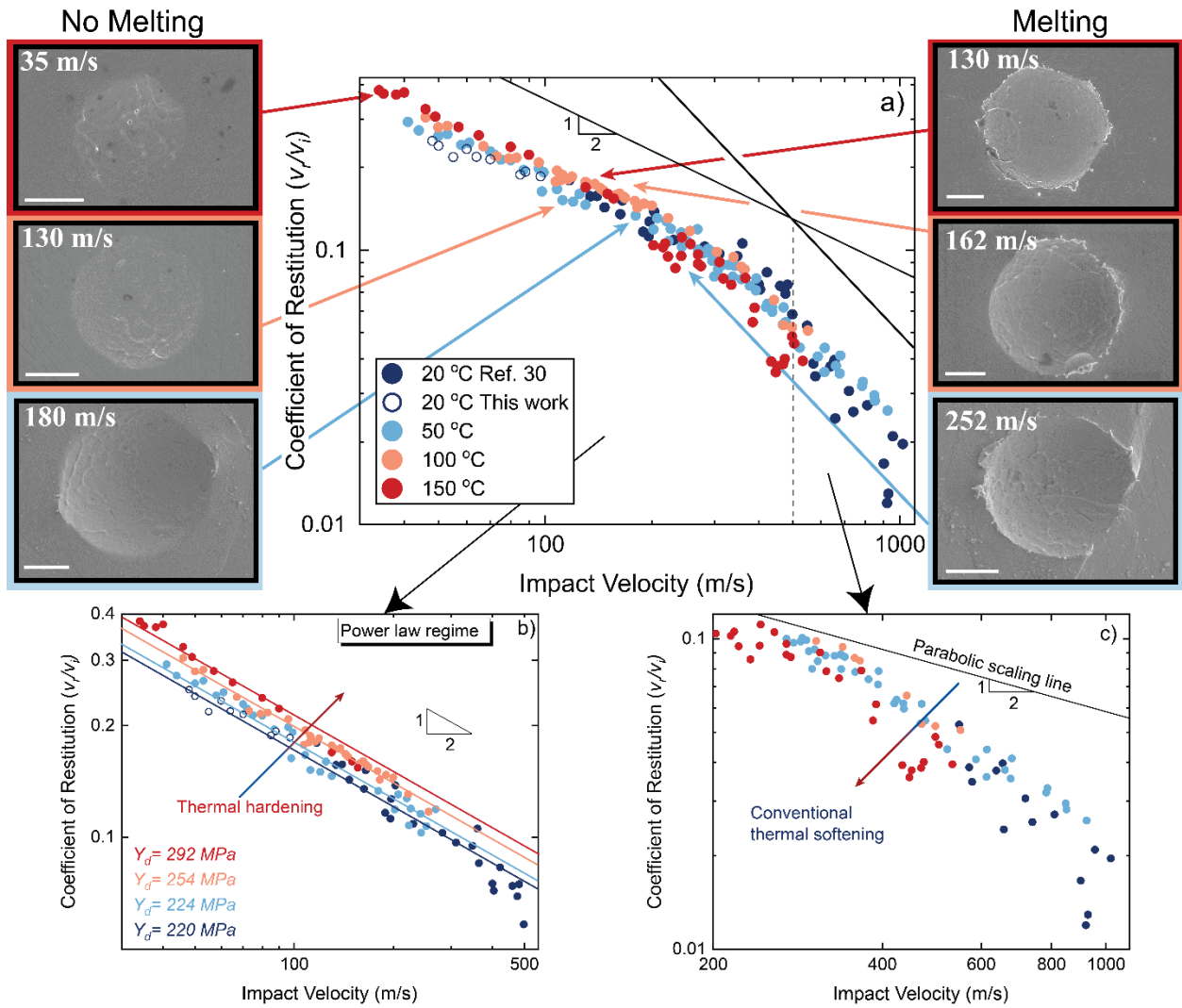


Figure 24: **a)** The data for each impact are plotted as the impact velocity versus coefficient of restitution (v_r/v_i) for each of the four temperatures, data for 20 °C reproduced from ref ³⁰. **b)** Only the impacts that follow the power law scaling relation are shown. Here a clear thermal hardening trend emerges from the data, seen as both faster rebounds at higher temperatures and the calculated dynamic yield strength values, Y_d . **c)** Impacts that diverge from the power law scaling, at faster impact velocities, are plotted. The thermal hardening trend appears to change back to conventional thermal softening as impacts at higher temperatures have slower rebound velocities. Impact craters from steel particle impacts are shown at three temperatures tested: 50 °C (blue), 100 °C (orange), and 150 °C (red). For each temperature, the impact craters show signatures of only plasticity and plasticity plus melting. The scale bar in each image is 3 μm .

The hardness of each impact is plotted in Fig. 25, divided into (a) pre-melting and (b) post-melting regimes. Interestingly, the boundary between the pre- and post-melting regimes aligns well with the divergence from parabolic scaling seen in Fig. 24. In the pre-melting regime, the craters form under constant volume constraint, whereas in the post-melting region, the crater volumes are larger due to the ejection of material, thus lowering the hardness. However, the temperature trends in Fig. 25 are striking: just as with the CoR data in Fig. 24, the value of H_d increases with increasing temperature in the pre-melting regime and decrease with increasing temperature in the post-melting regime. This can be most clearly seen when comparing the two temperature extremes, 20 and 150 °C, which are isolated in Fig. 25c. Just as with our earlier measurements of Y_d , we see here that at rates below $\sim 10^7 \text{ s}^{-1}$ antithermal hardening is observed before changing to conventional thermal softening above. Thus, two independent measurements agree about the mechanistic change from “hotter-is-stronger” to “hotter-is-softer” as the melting transition begins to manifest.

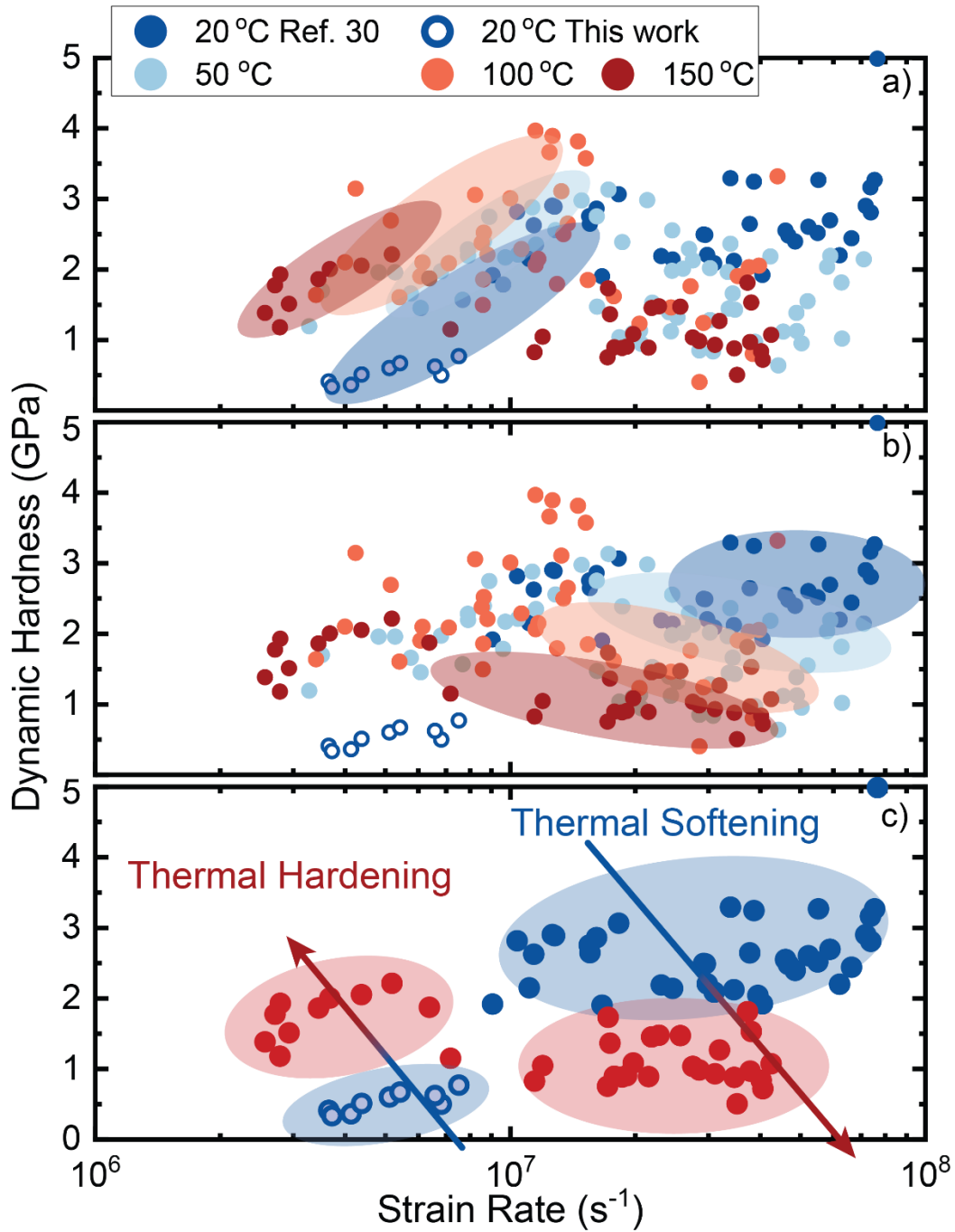


Figure 25: The dynamic hardness as a function of strain rate is plotted for each temperature in **a)** and the non-melting regime and **b)** the melting regime. The shaded regions are guides for the eye and highlight the data below, **a)**, and above **b)** the onset velocity for melting. The hardness trend in **a)** shows clear signs of thermal hardening with the hardness values at hotter temperatures being higher than at cooler temperatures. Past the point of substrate melting, **b)**, the trend in hardness changes where the hardness values at cooler temperatures are higher than at hotter temperatures. **c)** The hardness data for 20 °C and 150 °C are plotted separately from the other data, clearly highlighting the thermal hardening and thermal softening regimes.

4.2: Deformation Mechanisms Pre- and Post- Melting

At these high rates, for impacts that do not lead to substrate melting (i.e., the parabolic scaling regime in Fig. 24b), the strength of tin can be predicted using a mechanistic plasticity models such as commonly used to produce deformation mechanism maps^{19,25}. Antithermal hardening emerges in that regime due to the dislocation drag mechanism scaling linearly with temperature and dominating the other thermal (σ_{th}) and athermal (σ_a), terms that contribute additively to the total strength, Chapter 2:

$$\sigma_{tot} = \sigma_{th} + \sigma_a + \frac{3kT * (1.8b)}{5CV_c\rho_m b^2} \dot{\epsilon} \quad (30)$$

Both the intrinsic lattice strength σ_{th} , and the athermal, obstacle controlled (i.e., dislocation-dislocation and dislocation-grain boundary interaction) strength σ_a , decrease with temperature, due to increased thermal activation of dislocations over kinetic barriers, as well as a thermal decrease in the shear modulus. The last term in Eq. 30 represents phonon drag strengthening, where k is the Boltzmann constant, C is the bulk speed of sound, b is the magnitude of the burgers vector, ρ_m is mobile dislocation density, and V_c is the volume of the unit cell. In body centered tetragonal tin, the $\{100\}\langle 001 \rangle$ slip system with $\langle c \rangle$ burgers vector is the easiest for dislocation glide, and has the lowest lattice resistance of all of the available slip systems. Introducing the associated Burgers vector ($b = 0.159$ nm), along with the other materials properties of tin ($C = 3320$ m/s, $a = 0.583$ nm, $c = 0.318$ nm, $\rho_m \sim 10^{13} \text{ m}^{-3}$) and a proportionality

factor to convert from strength to hardness, we can generate a traditional deformation mechanism map as shown in Fig. 26. The regime of ballistic dislocation motion is captured in the lower-left hand corner, and shows contours of hardness that reflect antithermal hardening: “hotter-is-stronger” when this mechanism is dominant, and we see that the model’s contours are in good agreement with the experimentally measured hardness data, filled circles, in the power-law rebound regime where no melting occurs.

However, tin does not thermally harden indefinitely as the strain rate and temperature are increased, due to the onset of impact-induced melting from adiabatic heating^{30,31,36 96}. At a given test temperature, as the impact velocity is increased, melting first sets on and then increases in extent. It is intuitive that at higher test temperatures, less plasticity is needed to get to the melting point, and indeed, as the temperature is increased from 20 to 150 °C, the velocity where melting first occurs decreases dramatically from ~310 to ~75 m/s.

Impact melting such as this has been treated analytically by Hassani et al.³⁶ by mapping the kinetic energy of the impact to that needed to heat and melt a region near the surface; this gives a simple approximation for the velocity where melting is expected to set on, $v_{i,m}$:

$$v_{i,m} = \left[\frac{2\rho}{\alpha_i \beta m_p} \sqrt{\left(\frac{k}{\rho C_p} \right)} \cdot t_c A_c [C_p (T_m - T_o) + H_f] \right]^{1/2} \quad (31)$$

where $H_f = 60.2 \text{ J/g}$ is the latent heat of fusion, $C_p = 228 \text{ J/kg}\cdot\text{K}$ is the heat capacity, $\rho = 7265 \text{ kg/m}^3$ is the density of tin, $k = 60 \text{ W/m}\cdot\text{K}$ is the thermal conductivity, A_c is the contact area of the particle on the substrate, t_c is the contact time, $\beta = 0.6$, and $\alpha_i = 1$. This velocity for the onset of impact induced melting, can again be converted to a strain rate using the approximation that $v = \dot{\epsilon} \cdot d$. The onset of melting for each initial temperature is marked in Fig. 25a by the end of the shaded region. Interestingly, hardness data highlighted in Fig. 25a show antithermal hardening, while data beyond that zone (highlighted in Fig. 25b) show thermal softening, supporting the inference that the onset of substrate melting is the transition point from “hotter-is-stronger” to “hotter-is-softer”. Equation 31 can thus be used to directly chart the limit of the phonon drag regime on the deformation mechanism map of Fig. 26, which is presented with a thick black line. We are not aware of prior efforts to map this boundary, which reflects a shift to the dominance of deformation-induced melting and away from phonon drag. Past the melting boundary, the measured hardness is a superposition of the deformation mechanisms in Eq. 30 competing with fluid flow, which has the net effect of facilitating material ejection in our experiments. This ejection of material leads to a larger impact crater, beyond that of just a plastic indent in the surface.

In Fig. 26, above the onset of melting line, experimentally measured data show that increases in strain rate, temperature, or both lead to the classical “hotter-is-softer” trend. Dashed lines in the upper region act as a guide for the eye to show the trend in hardness data. In this region, measured crater volumes are increasing due to the loss of extra material upon impact beyond that of pile up from plasticity alone. Lienhard et al. has shown in Sn that when melting occurs and material is being ejected, the ejected material is both liquid and solid Sn, with a significant portion (> 90%) being solid.

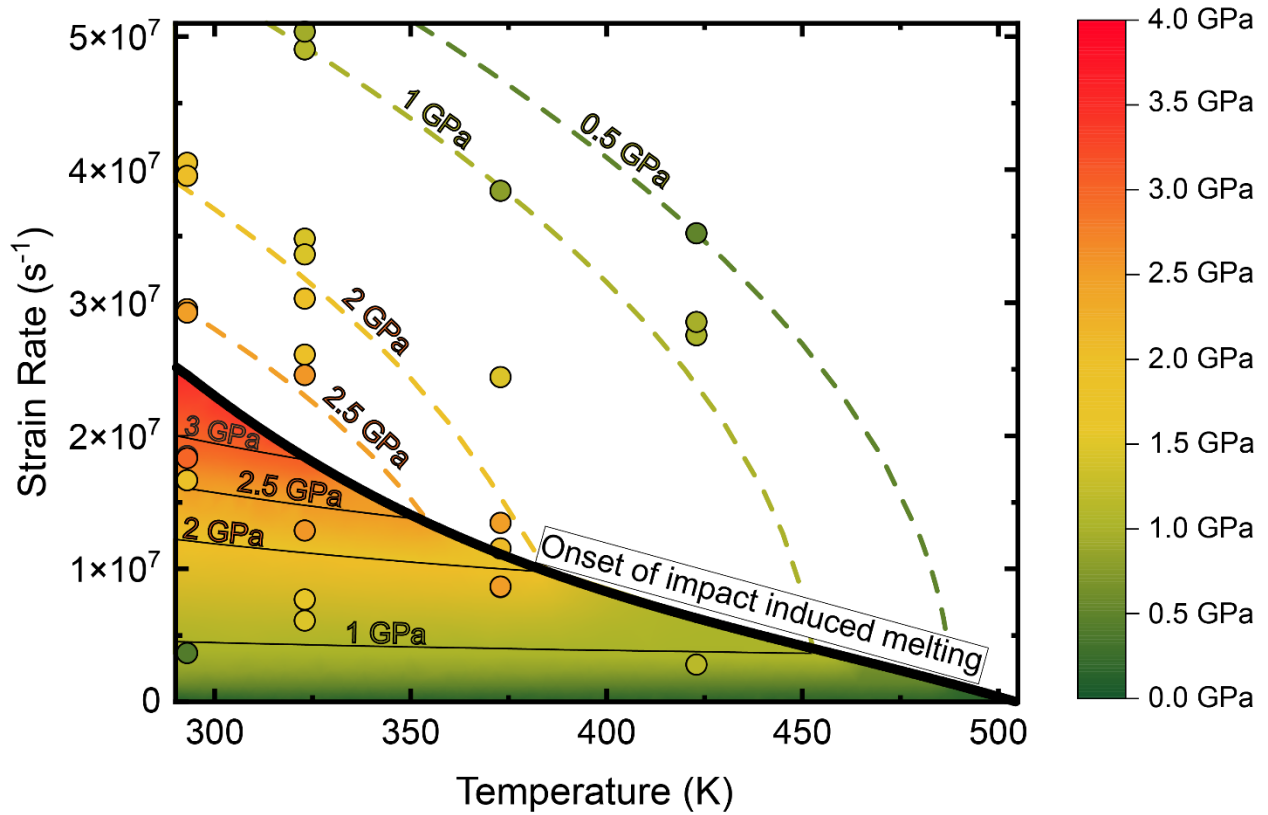


Figure 26: Hardness color contours as a function of temperature and strain rate are plotted for tin, up to the melting temperature (505 K). The thick black line indicates the boundary between melting and no melting for an impact under a given initial temperature and strain rate, and is the mechanistic transition point between thermal hardening (below) and thermal softening (above).

4.3: Summary

In summary, using tin as a model material for mechanical testing at strain rates above 10^6 s^{-1} and homologous temperatures above $0.80 T_m$, we find that there is an upper limit where the “hotter-is-stronger” phonon drag deformation mechanism prevails. This mechanism is bounded by the strain rate where impact induced melting from adiabatic heating first sets on for impacts

at a given test temperature. The development of liquid softens the metal and, in our experiments, leads to material ejection that directly lowers the apparent hardness. Once melting occurs, more conventional thermal softening, or “hotter-is-softer” behavior is observed. Because the phonon drag regime of deformation has been challenging to study with quantitative experiments, the ability of the present experimental method to map both this mechanism and its limits represents an opportunity for improved understanding of deformation mechanisms in extreme conditions.

Chapter 5: Conclusions and future work

Investigating the behavior and properties of materials at extreme strain rates, $>10^6 \text{ s}^{-1}$, has proved to be challenging due to the short time scales, on the order of tens to hundreds of nanoseconds, and the violent way in which these materials are deformed. Many common high-rate mechanical tests, including gas gun impact and Split-Hopkinson pressure bar experiments, are intrinsically limited to only testing materials up to strain rates of 10^5 s^{-1} before producing strong shock waves in the samples. Strong shock waves conflate the material response making it difficult to measure the mechanical behavior. Here, single microprojectile impact testing was used to assess the dynamic mechanical properties of metals at strain rates above 10^6 s^{-1} and well below the impact pressure where strong shock waves are generated.

In order to investigate the mechanical properties of materials, two independent measures of material strength were assessed, the dynamic yield strength and dynamic hardness. The dynamic yield strength is measured to be the average strength of the material over the entire duration of the impact event, and is fit from a plasticity model proposed by Wu et al.^{60,61}. This model averages the effects from all impacts that follow a parabolic scaling on the impact velocity verses coefficient of restitution plots. Complementary to the dynamic strength, the dynamic hardness for each impact experiment can be directly assessed as the plastic work, measured from the impact and rebound image sequence, and the impact crater volume, measured post impact with laser scanning confocal microscopy.

In Chapter 2, the first set of experiments to study the properties of metals at extreme rates was using alumina impactors on Cu, Ti, and Au substrates. Copper was chosen as a model

material because of its FCC crystal structure and its deformation behavior at low rates has been well studied. Impact experiments were carried out at three different temperatures, 20 °C, 100 °C, and 177 °C. For a constant impact velocity, as the temperature was increased, the rebound velocity of the particle was measured to be faster from the in-situ image sequences. This gave a first indication that the metals were stronger and harder with increasing temperature. Fitting this data to the plasticity model by Wu et al. showed that the dynamic yield strength of the material was increasing with increasing temperature, by ~30% from room temperature to 177 °C. Measured impact crater volumes supported this model, where for a constant impact velocity, hotter test temperatures led to smaller crater volumes, another indication of a harder material. Dynamic hardness measurements, showed a similar trend where impacts at hotter temperatures had a higher measured hardness. While this seemed counterintuitive at first, the anti-thermal behavior is the expected form of plasticity at these deformation rates. The strength of a metal can be approximated as the linear combination of the available strengthening mechanisms. At these rates the mechanisms are the thermally controlled strengthening, athermal strengthening, and dislocation-phonon drag strengthening. The dislocation-phonon drag strengthening term increases linearly with temperature, which leads to the total strength of the metals increasing strength with temperature. This is confirmed by calculating the apparent activation energy for plasticity from this strength model, which flips from positive to negative at strain rates of $\sim 10^5 \text{ s}^{-1}$ indicating a mechanistic change from thermally activated dislocation motion to ballistic transport of dislocations.

The work in Chapter 3 builds upon that of Chapter 2, but now looks to investigate the boundaries of where this anti-thermal “hotter-is-stronger” are. All of the metals used in Chapter 2 were high purity metals, however in many high-rate strength applications alloys instead of pure

metals are used. Solute elements in alloys act as additional pinning points for dislocations, and are a thermally activated strengthening mechanism. This time using five different compositions of Ni, ranging from 99.999% Ni to 99% Ni, impact experiments using alumina particles were done at two different test temperatures, 20 °C and 155 °C. As with the other pure metals, 99.999% Ni increased strength and hardness with increasing temperature. Interestingly, however, 99% Ni decreased strength with increasing temperature, pointing to the existence of a transition point, where there is a critical concentration of solute elements that change the mechanical response from thermal hardening to thermal softening. Starting with the strength model in Chapter 2, an additional strengthening term, solute strengthening, was added to now take into account changes in composition between the Ni samples. Plotting this model as a function of composition for two different temperatures, a cross-over point exists, $c \sim 0.4\%$, where the strength effect from solute elements is now greater than the contribution from dislocation-phonon interactions.

Chapter 4 turns back to the effect of temperature in pure metals, namely tin. While it was shown that metals increase strength with increasing temperature, this cannot happen indefinitely. One upper bound of this is the melting temperature of the material. By using tin, it is easy to reach high homologous temperatures at moderate absolute temperatures and extreme strain rate impacts. At four different temperatures 20 °C, 50 °C, 100 °C, and 150 °C, properties of tin were assessed for impacts of steel particles at impact velocities ranging from ~ 30 m/s to above 1000 m/s. For a constant temperature, as the impact velocity increased, there was a clear transition from only plasticity to plasticity and melting from adiabatic heating at the impact site. As the initial test temperature increased, this transition point occurred at lower velocities. More interestingly, the dynamic hardness increased with increasing temperature for impacts that did not lead to substrate melting, but decreased with increasing temperature when substrate

melting was present. Melting of substrate material allowed for material to be ejected from the impact crater beyond that of just plasticity. This led to a larger impact crater which gives rise to the decreased measured dynamic hardness.

Looking forward, there is an abundance of work still to be done when studying metals at extreme strain rates. Much of the work up to now has dealt with pure metals, however there is much interest in going beyond pure metals. Investigating the extreme rate properties of dual phase materials using single microprojectile impact testing is attractive because the two different phases can be investigated independently and together to understand how each phase influences the overall mechanical response of the metal.

Further, there has been little work done to experimentally capture material that is ejected from an impact crater. In the case of tin, it has been predicted that most of the ejected material is solid and not molten, however this material has not been captured post impact before. By capturing this material, the amount of energy partitioned between plasticity, rebound, and substrate melting can be refined and the mechanism for why the amount of material that is ejected can be studied in-situ. In some instances of higher melting temperature metals, material is ejected from the impact crater, however no signatures of melting are seen in post impact micrographs.

Finally, to date, impact experiments using microprojectile impact testing have only been carried out at room and elevated temperatures, but there is interesting physics that happen when going lower in temperature. At temperatures below room temperature, the dislocation-phonon drag mechanism begins to become insignificant, However, a new strengthening mechanism becomes active that isn't seen at higher temperatures, the dislocation-electron drag

mechanism. This mechanism is quantitatively underexplored using dynamic hardness measurements, however mapping the entire temperature space for a metal such as pure Cu would be advantageous. The transition between dislocation-electron drag and dislocation-phonon drag could, for the first time be seen using a dynamic mechanical test and modeled using a similar strength model to those seen in Chapters 2-4.

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